



Dr. Vishwanath Karad

**MIT WORLD PEACE
UNIVERSITY** | PUNE

TECHNOLOGY, RESEARCH, SOCIAL INNOVATION & PARTNERSHIPS

Network and Cyber Security (AID4PM04A)

Department of Computer Engineering and Technology (SCET)

COURSE STRUCTURE

Course Code	AID4PM04A				
Course Category	Program Major				
Course Title	Network and Cyber Security				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	2	--	2	--	4
Credits	2	--	1	--	3
Assessment Schema Code	TL4				

Pre-requisites

- Programming and Problem Solving
- Principles of Programming Languages with Python Programming
- Object Oriented Programming with C++

Course Objectives:

By participating in and understanding all facets of this Course a student will be able:

Knowledge:

- i. To understand the fundamentals of networking and data communications and fundamental concepts and principles of information security and cyber threats.
- ii. To understand basics of physical layer, data link layer, network layer and transport layer.
- iii. Familiarity with security frameworks, protocols, and best practices for protecting digital assets.

Skill:

- i. To learn network layer protocols and identify different services offered by transport layer.
- ii. Ability to identify and mitigate security vulnerabilities and threats in various systems.
- iii. Proficiency in implementing and managing security policies, practices, and technologies to protect information systems.

Attitude:

- i. Commitment to continuous learning and staying updated with evolving concepts of networking, application areas, classification, reference models, transmission environment, technologies in routing algorithms and recent security trends and technologies.
- ii. Responsibility towards ensuring ethical practices and maintaining the confidentiality, integrity, and availability of information.

Course Outcomes:

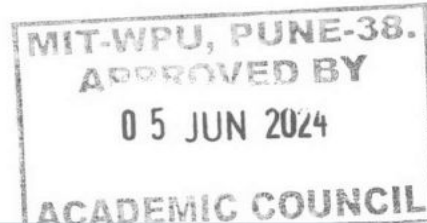
On completion of course, students will be able to:

1. Identify characterizing features of transmission techniques.
2. Recall and Demonstrate working of flow control & error control protocols.
3. Define connection oriented & connectionless transport layer protocols & build working applications based on it.
4. Explain the core principles of information security and cryptographic techniques, including their relevance and application to modern security challenges.
5. Apply number theory concepts to implement and test basic public key encryption and decryption algorithms.

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6. Demonstrate the ability to configure network security tools and implement/examine security techniques to monitor and defend different against cybercrime.

Syllabus

Unit I Fundamentals of networking and data communication

Introduction to data communication, Transmission modes (parallel and serial), Digital Transmission: Digital-to-Digital conversion (line coding, block coding), Analog-to-Digital conversion (Pulse Code Modulation, Delta Modulation)

Introduction to computer networks, Applications of computer networks, Types of Networks, Network Architectures, Network Topologies, Transmission media, Networking devices, OSI Model, TCP/IP Reference Model, Ethernet standards.

Unit II Data Link Layer

Error Detection, Error Correction, Elementary Data Link Layer Protocols, Sliding Window Protocols: Stop and Wait ARQ, Go back N ARQ, Selective Repeat, Medium Access Layer, Channel Allocation Problem, Multiple Access Protocols, Network Layer Design Issues, Routing Algorithms, Congestion Control Algorithm.

Unit III Network Layer

Network Layer: IP Protocol, IP Address, Internet Control Protocol, Transport Layer Design Issues, Connection management, Addressing, Establishing and releasing connection, simple transport protocol, Internet transport protocol.

Unit IV Network Security and Security Practices

Overview, Symmetric Cipher: Classical Encryption Techniques, Symmetric Cipher: Block and Data Encryption Standards, Public Key Encryptions, Hash Functions, Public key cryptography and RSA, Network Security Practices, Authentication Applications

Guidelines for CCA and LCA

Examination Scheme

Sr. No.	Examination Scheme	Marks
1.	Class Continuous Assessment (CCA)	40
2.	Laboratory Continuous Assessment (LCA)	20
3.	End Term Examination	40

Sr. No.	Laboratory Continuous Assessment: (20 Marks)	Marks
1.	LCA1-Practical Performance	5
2.	LCA2-Active learning / Additional implementation	10
3.	LCA3-End Term Practical/ Oral Examination	5

CCA Marks Distribution

Sr. No.	Class Continuous Assessment: (40 Marks)	Marks
1.	CCA1-poster making ,MCQ	10
2.	Mid Term Exam	20
3.	CCA2-online course(above 20 hrs)	10

Lab Assignments

Lab Contents

1. Design Simple Network using different topology (Star, Mesh, Bus, Extended star etc.): (Use: Packet Tracer/GNS)
2. Implement Error Control Protocol (error detection and correction Hamming code): (Language: C/C++/Python/Java)
3. Write a program to demonstrate Sub-netting and find subnet masks
4. Write a program for TCP/UDP Socket Programming: (Language: C/C++/Python/Java).
5. Write a program using JAVA or Python or C++ to implement simplified data encryption algorithm (S-DES)
6. Write a program using JAVA or Python or C++ to implement RSA
7. Configuration and demonstration of Intrusion Detection System using Snort

Learning Resources:

Text Books:

1. Behrouz A. **Forouzan**, 'Data Communications and Networking', 5th Edition, McGrawHill Publishing Company, ISBN 978-0-07-337622-6
2. **Tanenbaum** A. S., 'Computer Networks', Pearson Education, 5th Edition, ISBN-978-0-13-212695-3
- 3 William **Stallings**, Cryptography and Network Security: Principles and Practice, Pearson Publication
- 4 Nina Godbole & Sunit Belapure, Cyber Security: Understanding Cybercrimes, Computer Forensics and Legal Perspectives, Wiley Publication
- 5 K. Pachghare, Cryptography and Information Security, PHI Learning

Learning Resources:

Reference Books:

1. James F. Kurose and Keith W Ross 'Computer Networking, A Top-Down Approach', 5th Edition, Pearson Education, ISBN- 978-81-317-9054-0
2. W. Richard Stevens, Unix Network Programming, The Sockets Networking API, Vol 1, 3rd Edition, PHI Learning Pvt. Ltd.
3. Atul Kahate, "Cryptography and Network Security", McGraw Hill Education
4. Earnest A. Kallman, J.P Grillo, "Ethical Decision making and IT: An Introduction with Cases", McGraw Hill Publication.
5. Charles A. Sennewald, Curtis Baillie, "Effective Security Management", Elsevier Publication

Supplementary Reading:

1. William Stallings, 'Data and Computer Communications', 6th Edition, Prentice Hall of India Pvt.
2. Michael E. Whitman and Herbert J. Mattord , "Principles of Information Security", Cengage Learning; ISBN: 1285448367
3. Christ of Paa and Jan Pelzl, "Understanding Cryptography: A Textbook for Students and Practitioners", Springer: ISBN: 3642041000

Learning Resources:

Web Resources:

1. <https://nptel.ac.in/courses/106/105/106105080/>
2. <https://nptel.ac.in/courses/106/106/106106091/>
3. <https://www.newhorizons.com/promotions/cybersecurity-ebooks>

Web links:

1. <https://nptel.ac.in/courses/106/105/106105081/>
2. <https://nptel.ac.in/courses/106/105/106105183/>
3. <https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-829-computer-networks-fall-2002/index.htm>

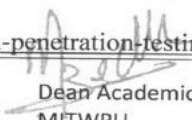
MOOCs:

1. <https://www.coursera.org/learn/computer-networking>
2. <https://www.edx.org/course/introduction-to-networking?index=product&queryID=0befdc538866babf2facfdc2dce06204&position=1>
3. <https://nptel.ac.in/courses/106106129>
4. <https://www.udemy.com/course/hands-on-penetration-testing-labs-30/>

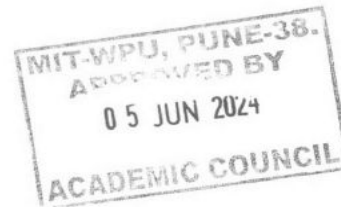
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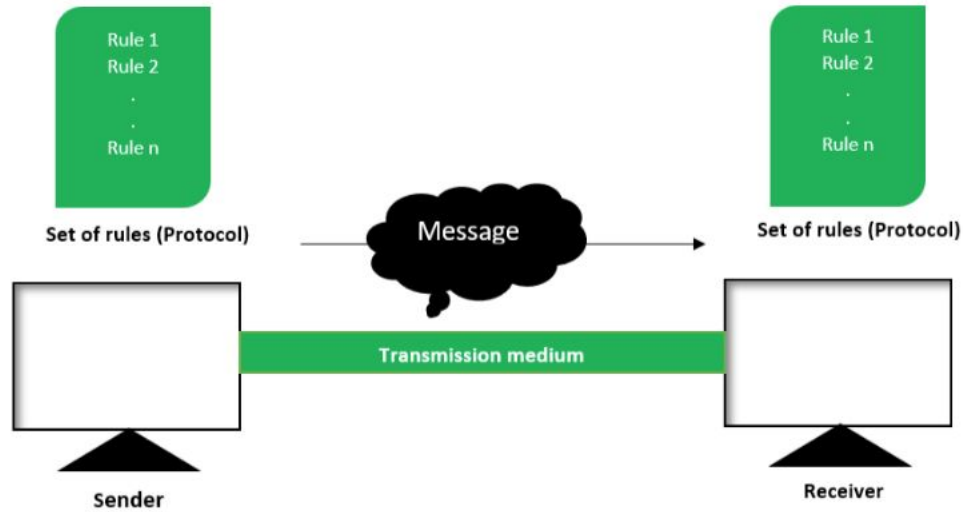


Unit I: Fundamentals of Networking and Data Communication

- **Introduction to data communication:** Transmission modes(parallel and serial)
- **Digital Transmission:** Digital-to-Digital conversion (line coding, block coding), Analog-to Digital conversion (Pulse Code Modulation, Delta Modulation)
- Introduction to computer networks, applications of computer networks
- **Types of Networks,** Network Architectures, Network Topologies,
- **Transmission media,** Networking devices, OSI Model, TCP/IP Reference Model,
- **Ethernet standards.**

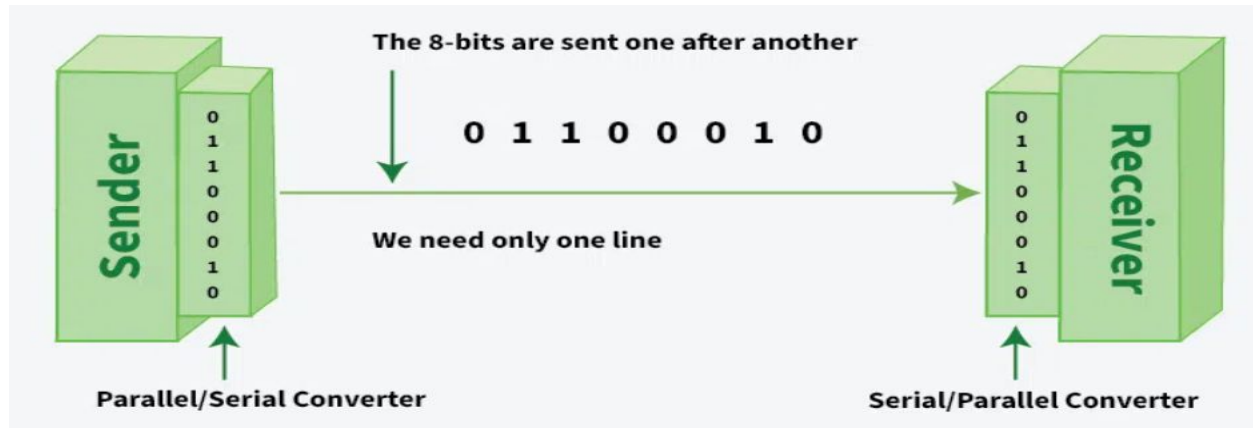
Introduction

- ❑ **Data Communication** - Exchange of data between two devices via some form of transmission medium.
- ❑ It refers to the process of transmitting and receiving data between two or more devices over a communication channel. It involves the conversion of data into signals that can be transmitted and then decoding those signals at the receiving end.
- ❑ **Data** is a collection of facts. **Information** is how you understand those facts in context.

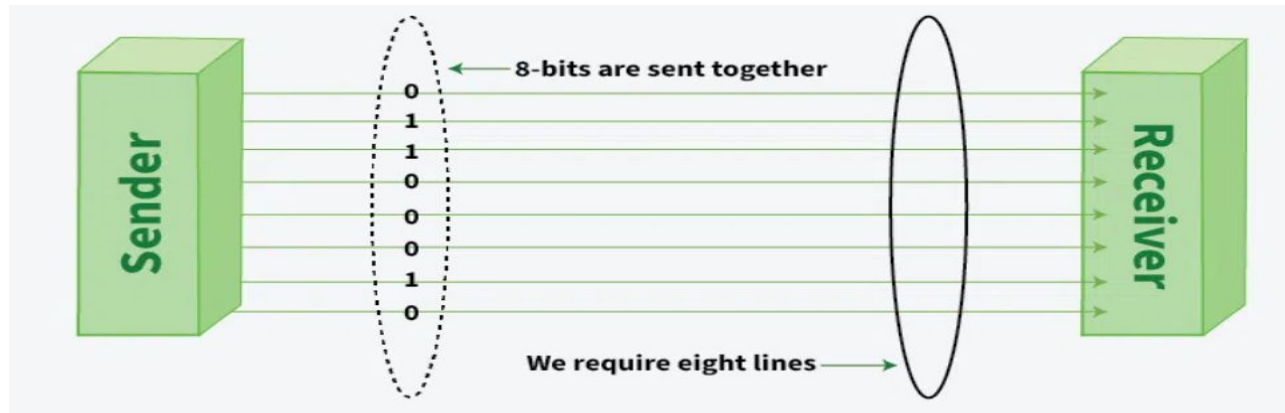


Serial and Parallel Transmission

In computer networks for data communication the sending or receiving devices use parallel transmission internally, converters (serial to parallel and parallel to serial) are used at the interface point between the device and the line.

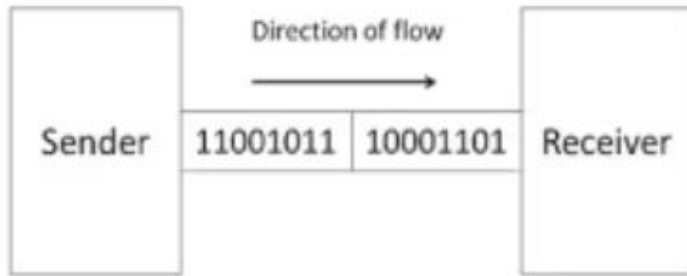


Serial transmission

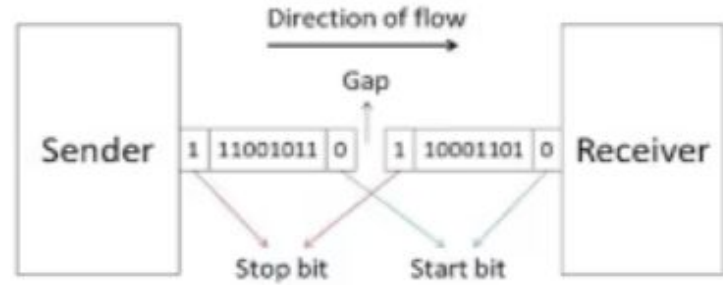


Parallel Transmission

- In **Serial transmission**, data is sent bit by bit from one computer to another using a two-way path.
- Each bit has its own timing.
- 8 bits are sent at a time, with extra bits at the start and end (called Parity bits), which are 0 and 1.
- This method uses special cables to send data over long distances.
- The data is sent in order.



Synchronous Mode



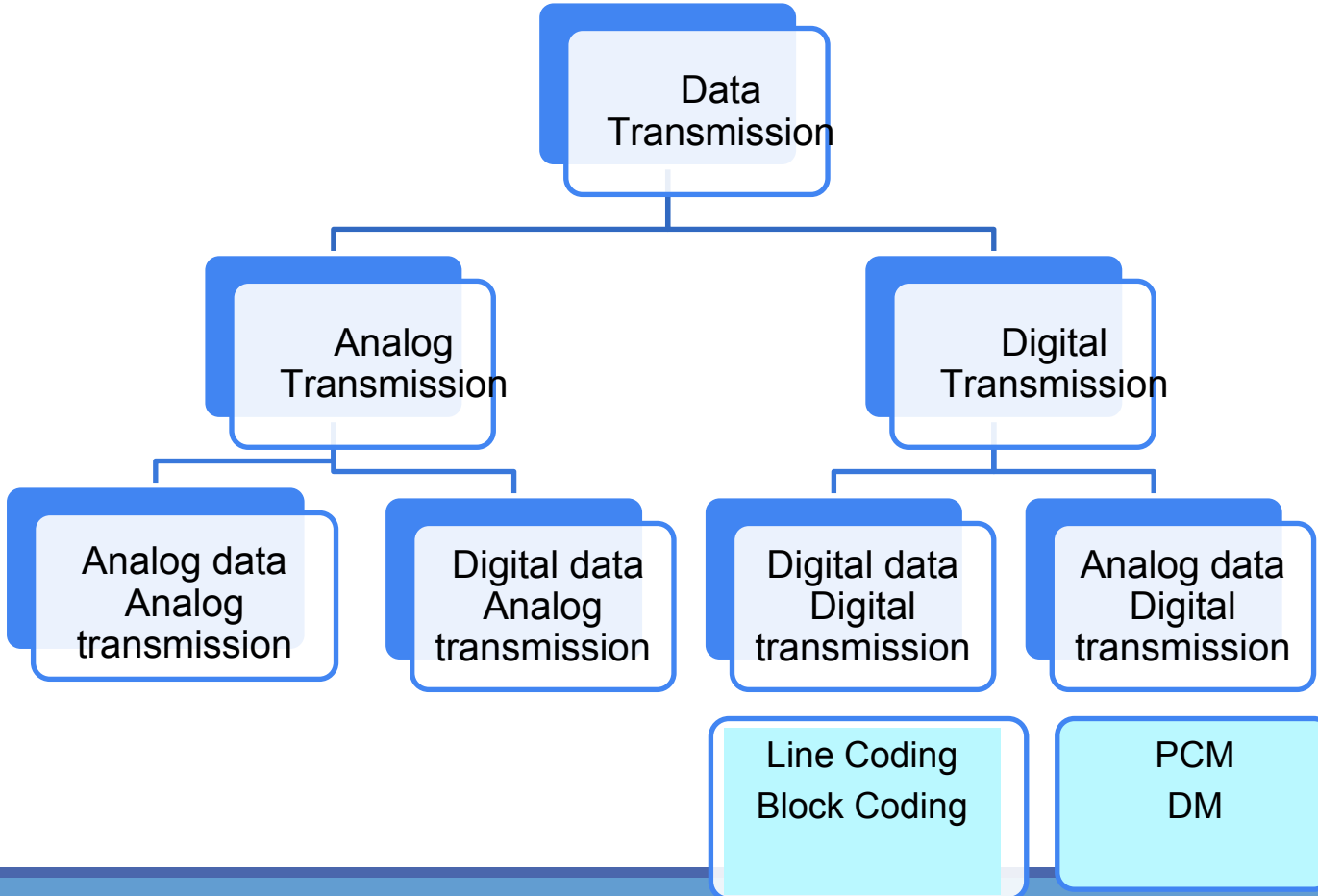
Asynchronous Mode

- **Parallel Transmission** sends many bits of data at the same time from one computer to another.
- It's faster than Serial and works well for short distances.
- It fits well with how computers are built inside, as they use parallel systems.
- Computers and other devices can easily use Parallel Transmission.

Serial Transmission	Parallel Transmission
In this type, a single communication link is used to transfer data from one end to another	In this type, multiple parallel links are used to transmit the data
In serial transmission, data(bit) flows in bi-direction.	In Parallel Transmission, data flows in multiple lines.
Serial Transmission is cost-efficient.	Parallel Transmission is not cost-efficient.
In serial transmission, one bit is transferred at one clock pulse.	In Parallel Transmission, eight bits are transferred at one clock pulse.
Serial Transmission is slow in comparison of Parallel Transmission.	Parallel Transmission is fast in comparison of Serial Transmission.

Generally, Serial Transmission is used for long-distance.	Generally, Parallel Transmission is used for short distance.
The circuit used in Serial Transmission is simple.	The circuit used in Parallel Transmission is relatively complex.
Serial Transmission is full duplex as sender can send as well as receive the data	Parallel Transmission is half-duplex since the data is either send or receive
Converters are required in a serial transmission to convert the data between internal and parallel form	No converters are required in Parallel Transmission
Serial transmission is reliable and straightforward.	Parallel transmission is unreliable and complicated.

Data Transmission

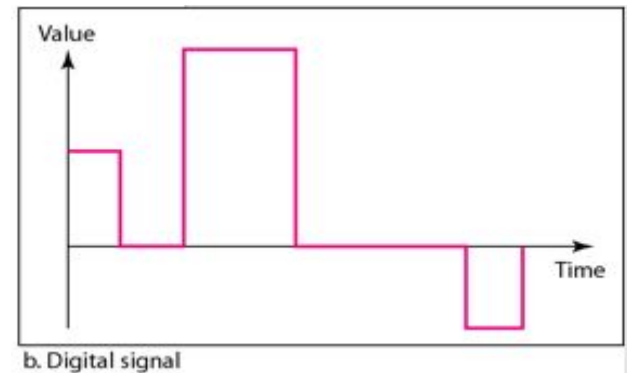
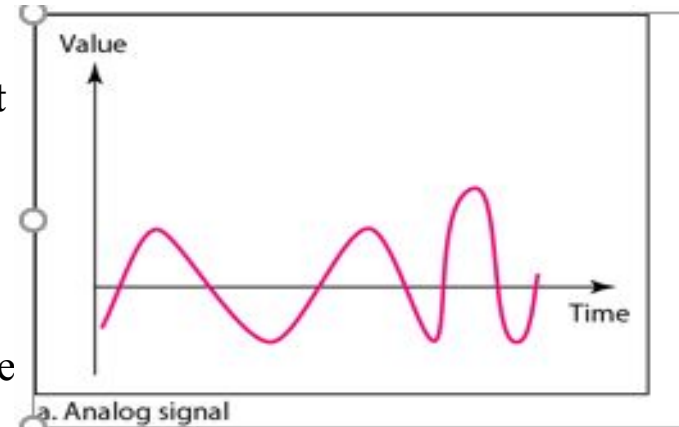


Analog and Digital Data

- ❑ The term analog data refers to information that is **continuous**.
- ❑ Analog data, such as the **sounds made by a human voice**, take on continuous values can be captured by a **microphone** and converted to an analog signal or sampled and converted to a digital signal.
- ❑ **Digital data** refers to information that has **discrete states**.
- ❑ **Digital data** take on discrete values. For example, data are stored in the memory as **0s and 1s**.
- ❑ They can be converted to a digital signal or modulated into an analog signal for

Analog and Digital Signals

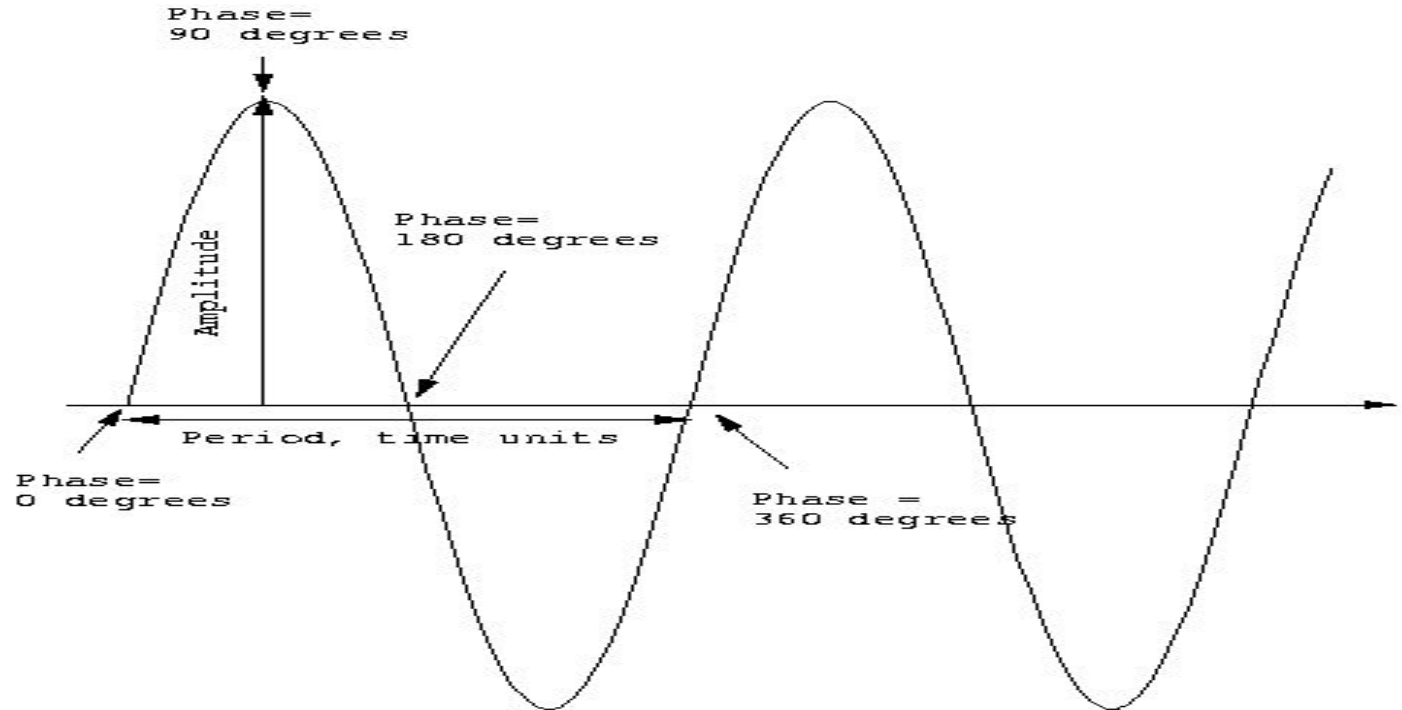
- **Signal** - A signal is an **electromagnetic or electrical current** that is used for carrying data from one system or network to another.
- **An analog signal** is any **continuous signal** for which the time varying feature of the signal is a representation of some other time varying quantity i.e., analogous to another time varying signal.
- **A digital signal** is a signal that represents data as a **sequence of discrete values**; at any given time it can only take on one of a finite number of values.



Difference between Analog and Digital Signals

Analog Signals	Digital Signals
Analog signal is continuous and time varying .	Digital signal have two or more states and in binary form .
Troubleshooting of analog signals are difficult .	Troubleshooting of digital signals are easy .
An analog signal is usually in the form of sine wave .	An digital signal is usually in the form of square wave .
Easily affected by the noise .	These are stable and less prone to noise.
Analog signals use continuous values to represent the data.	Digital signals use discrete values to represent the data.
Accuracy of the analog signals may be affected by noise.	Accuracy of the digital signals are immune from the noise.
Analog signals may be affected during data transmission .	Digital signals are not affected during data transmission.
Analog signals use more power .	Digital signals use less power .
Examples: Human voice, natural sound , Audio signal, Radio signal, light, temperature etc.	Examples: Computers, Optical drives, etc.
Components like resistors, Capacitors, Inductors are used in analog circuits.	Components like transistors, logic gates, and micro-controllers are used in Digital circuits.

Characteristics of Waveform



- Modulation is the process of altering **one or more** properties of a periodic waveform known as a **carrier signal** with respect to **the modulating signal**, which contains **information** to be transmitted.
- Modulation is performed by the device known as **a modulator**, and this technique is mainly used to **overcome** the interference of the signal.
- **Modulation techniques** typically aid in **long-distance communication**.
- Modulation is of two types:

1. Analog Modulation

Digital-to-Digital Conversion

1. Line coding
2. Block coding

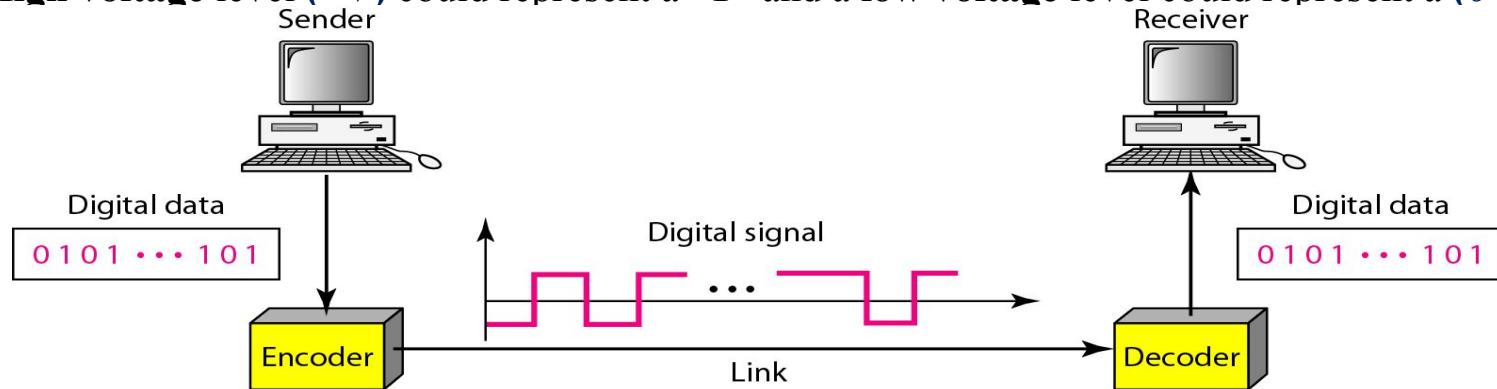
Digital-to-Digital conversion

- **Digital-to-digital conversion** techniques, methods which convert digital data to digital signals.
- The conversion involves **techniques**:
 1. **Line Coding**
 2. **Block Coding**

Note: Line coding is always needed; block coding may or may not be needed.

Line Coding

- **Line coding** is the process of converting digital data to digital signals.
- **At the sender**, digital data are **encoded** into a digital signal; **at the receiver**, the digital data are recreated by **decoding** the digital signal.
- **High voltage level (+V)** could represent a “1” and a **low voltage level** could represent a **(0 or -V)**.



Data Rate Versus Signal Rate

- The **data rate** defines the number of data elements (bits) sent in 1s.
- The unit is **bits per second (bps)**.
- The **signal rate** is the number of signal elements sent in 1s. The unit is the baud.
- There are several common terminologies used in the literature.
- The **data rate** is sometimes called the **bit rate**; **the signal rate** is sometimes called the **pulse rate, the modulation rate, or the baud rate**.
- One goal: **To increase the data rate while decreasing the signal rate.**

Line Coding Characteristics

1. Data Rate versus Signal Rate:

- The **data rate** defines the **number of data elements (bits) sent in 1s**. The unit is bits per second (bps). The data rate is sometimes called the **bit rate**
 - The **signal rate** is the **number of signal elements sent in 1s**. The unit is the baud. The signal rate is sometimes called the pulse rate, the modulation rate, or the **baud rate**.
 - **Goal in data communications is to increase the data rate while decreasing the signal rate.**
- Increasing the **data rate** increases the **speed of transmission**.
 - Decreasing the **signal rate** decreases the **bandwidth requirement**.

Line Coding Characteristics

2. *Bandwidth:*

- Digital signal that carries information is **non-periodic**.
- The **bandwidth of a non-periodic signal is continuous with an infinite range**.
- However, most digital signals we encounter in real life have a bandwidth with finite values.
- The bandwidth is theoretically infinite, but many of the **components have such a small amplitude that they can be ignored**.
- **The effective bandwidth is finite.**

Line Coding Characteristics

3. *Baseline Wandering:*

In decoding a digital signal, the receiver calculates a **running average** of the received signal power. This average is called the **baseline**.

- The incoming signal power is evaluated against this baseline to determine the value of the data element.
- A long string of 0 s or 1s can cause a drift in the baseline (baseline wandering) and make it difficult for the receiver to decode correctly.
- A **good line coding scheme** needs to **prevent baseline wandering**.

Line Coding Characteristics

5. *DC Components:*

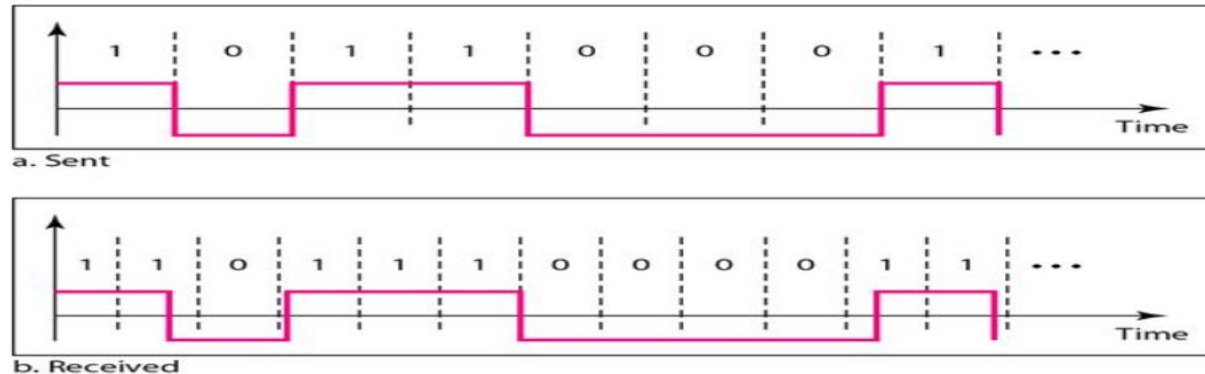
- When the voltage level in a digital signal is constant for a while, the spectrum creates **very low frequencies**
- These frequencies around zero, called DC (direct-current) components
- This DC component present problems for a system that **cannot pass low frequencies** or a system that uses electrical coupling (via a transformer).

Line Coding Characteristics

6. Self-synchronization:

- To correctly interpret the signals received from the sender, **the receiver's bit intervals must correspond exactly to the sender's bit intervals.**
- If the receiver clock is faster or slower, the bit intervals are not matched and the receiver might misinterpret the signals.

The following figure represents the synchronization problem.



Line Coding Characteristics

7. Built-in Error Detection:

It is desirable to have a built-in error-detecting capability in the generated code to detect some of or all the errors that occurred during transmission.

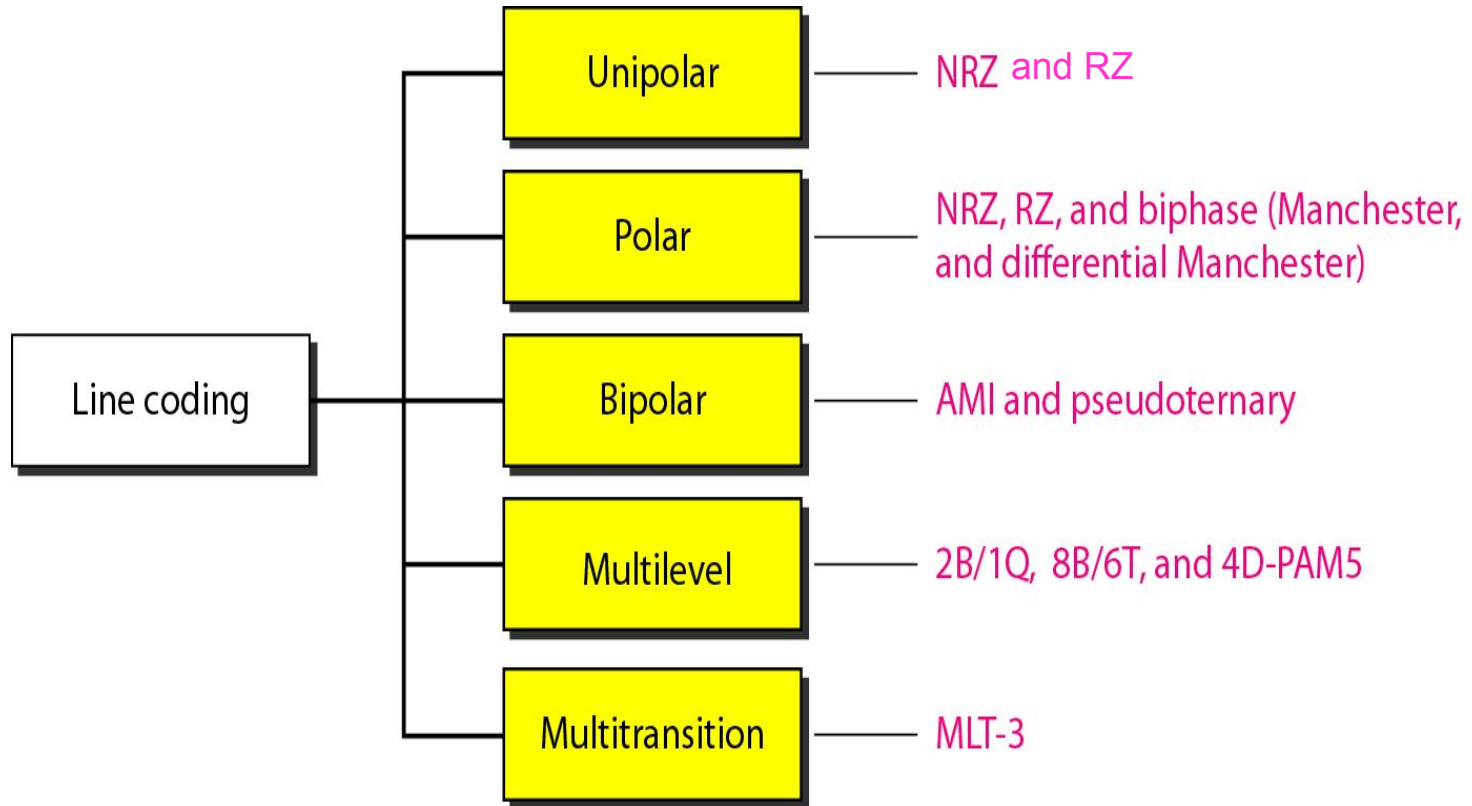
8. Immunity to Noise and Interference:

Another desirable code characteristic is a code that is immune to noise and other interferences.

9. Complexity:

A complex scheme is more costly to implement than a simple one. For example, a scheme that uses four signal levels is more difficult to interpret than one that uses only two levels.

Line Coding Schemes

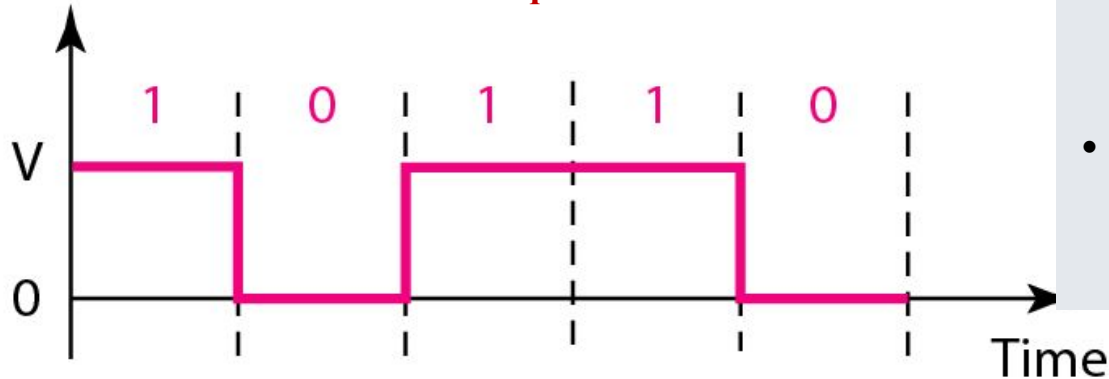


1. Unipolar NRZ scheme

- In a unipolar scheme, all the signal levels are on **one side** of the time axis, either above or below. In **Non-Return-to-Zero**, the signal does not return to zero at the middle of the bit, *where* positive voltage defines

Amplitude

Sequence of bits=10110



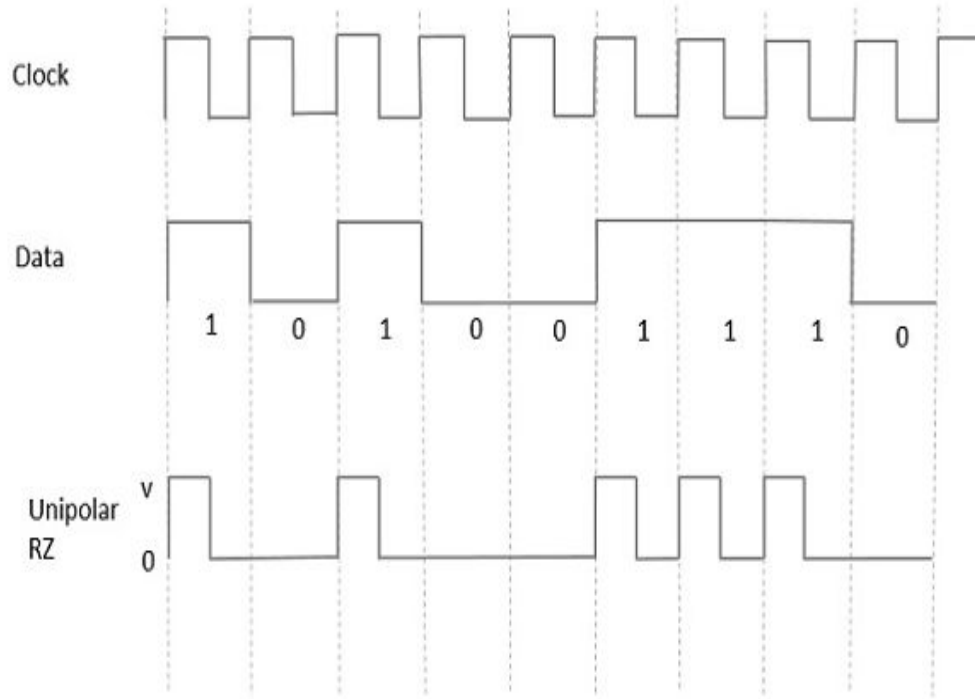
Advantages

- It is simple.
- A lesser bandwidth is required.

Disadvantages

- No error correction done.
- Presence of low frequency components may cause the signal droop.
- Loss of synchronization is likely to occur (especially for long strings of 1s and 0s).

1. Unipolar Return to Zero



Advantages

- It is simple.
- The spectral line present at the symbol rate can be used as a clock.

Disadvantages

- The disadvantages of Unipolar RZ are –
- No error correction.
- Occupies twice the bandwidth as unipolar NRZ.
- The signal droop is caused at the places where signal is non-zero at 0 Hz

2. Polar NRZ scheme

a. Non-Return-to-Zero (NRZ):

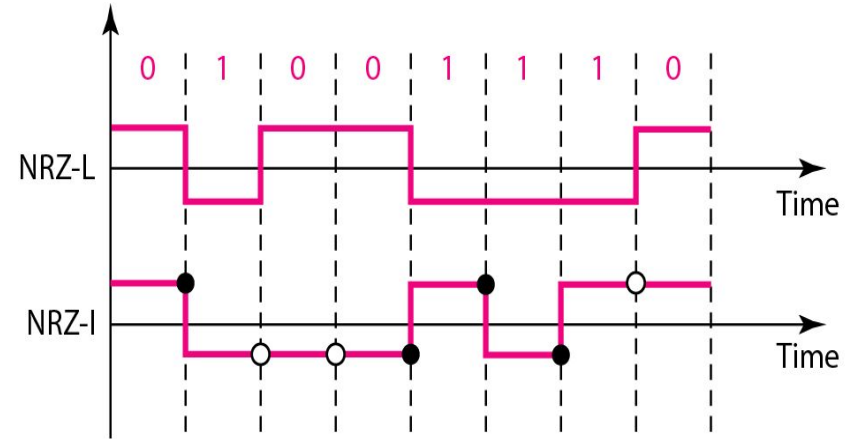
The voltages are on both sides of the time axis.

Polar NRZ scheme can be implemented with **two voltages**. E.g. $-V$ for 1 and $+V$ for 0.

There are **two versions**:

1. NZR - Level (NRZ-L) - positive voltage for zero symbol and negative for the other.

2. NRZ - Inversion (NRZ-I) - the change or lack of change in polarity determines the value of a symbol. E.g. a “1” symbol inverts the polarity a “0” does not.



○ No inversion: Next bit is 0 ● Inversion: Next bit is 1

Advantages

- It is simple.
- No low-frequency components are present.

Disadvantages

- No error correction.
- No clock is present.
- The signal droop is caused at the places where the signal is non-zero at **0 Hz**.

2.Polar RZ scheme

b. Return-to-Zero (RZ):

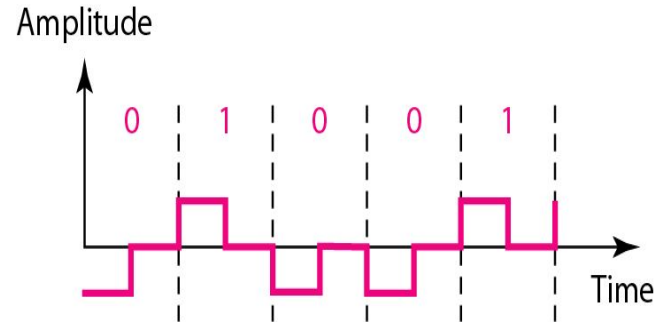
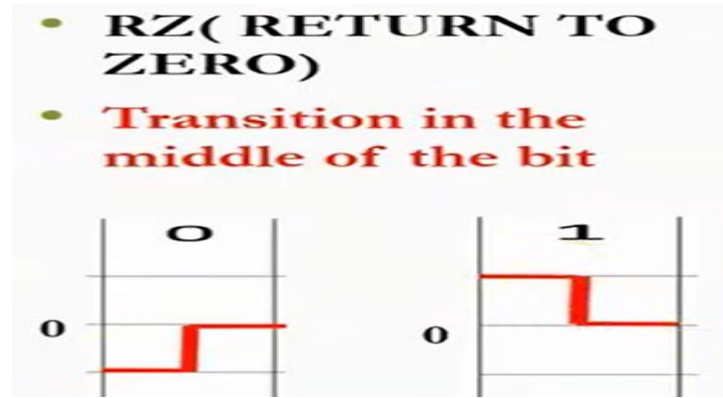
- This scheme uses **three voltage values. +, 0, -.**
- Each symbol has a **transition in the middle.**

Advantages

- It is simple.
- No low-frequency components are present.

Disadvantages

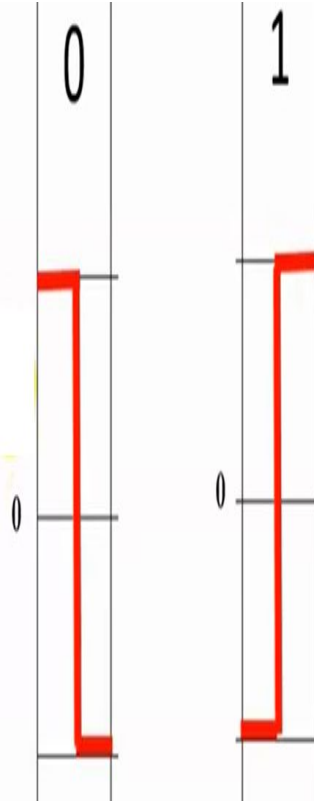
- No error correction.
- No clock is present.
- Occupies twice the bandwidth of Polar NRZ.
- The signal drop is caused at places where the signal is non-zero at **0 Hz**.



2. Polar Schemes : Manchester

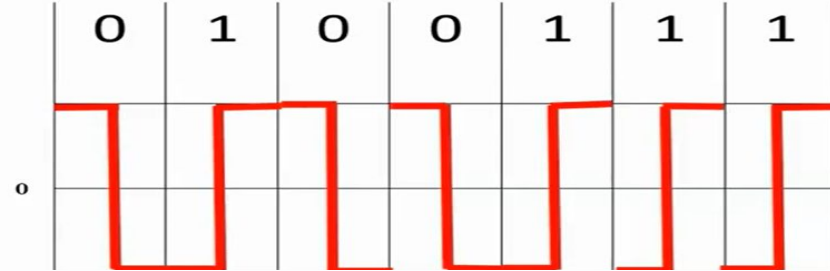
BIPHASE

- MANCHESTER
- Combines the logic of RZ and NRZ-L
- 0: Positive to negative transition
- 1: Negative to positive transition



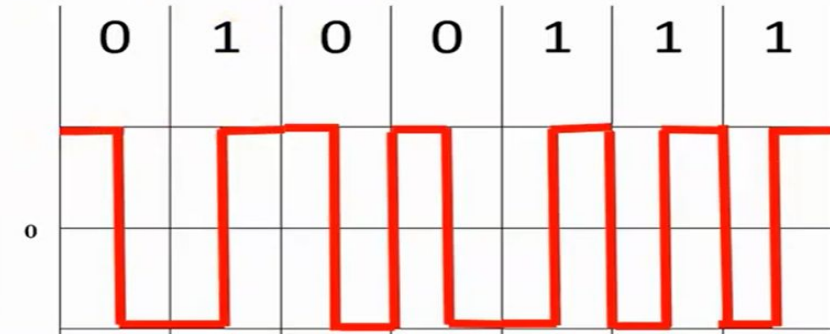
Example 1:

Data : 0 1 0 0 1 1 1



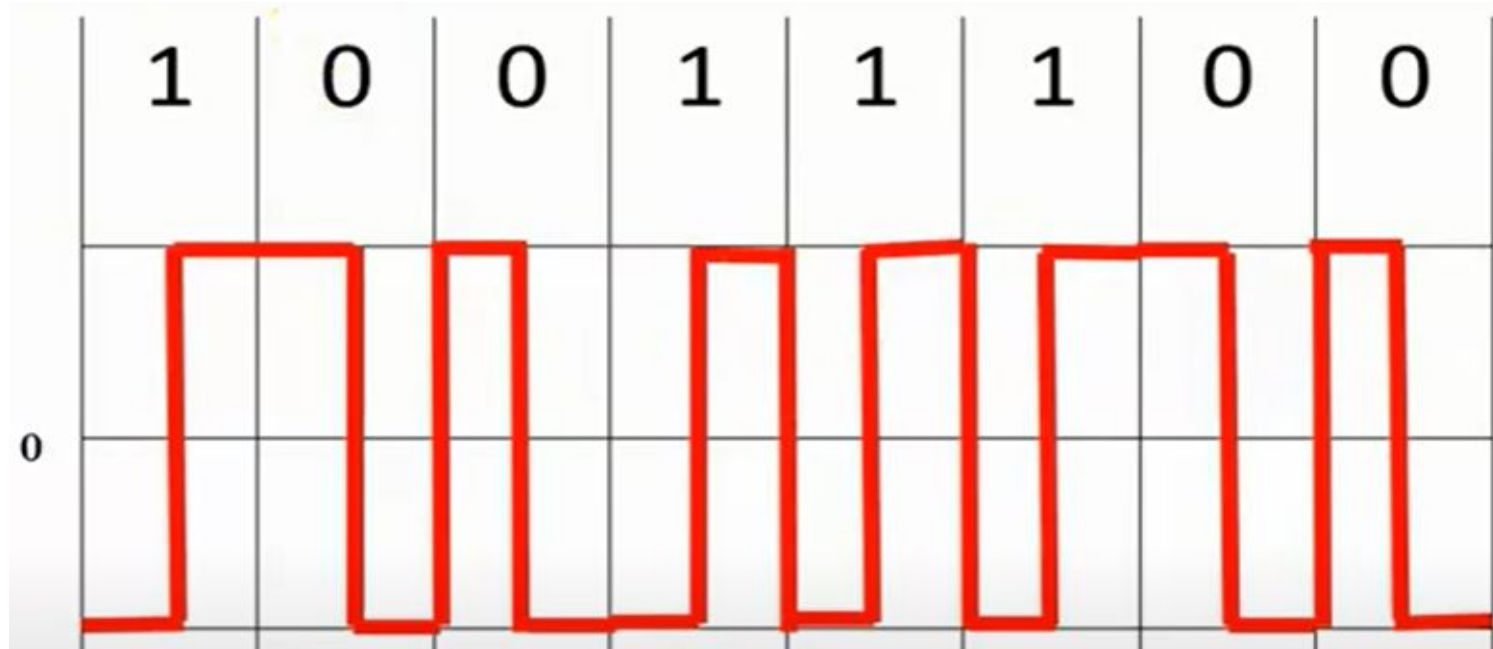
Example 1:

Data : 0 1 0 0 1 1 1



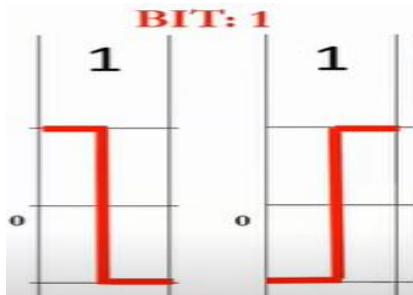
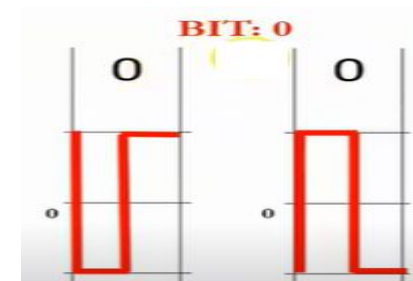
2. Polar Schemes : Manchester

Data : 1 0 0 1 1 1 0 0

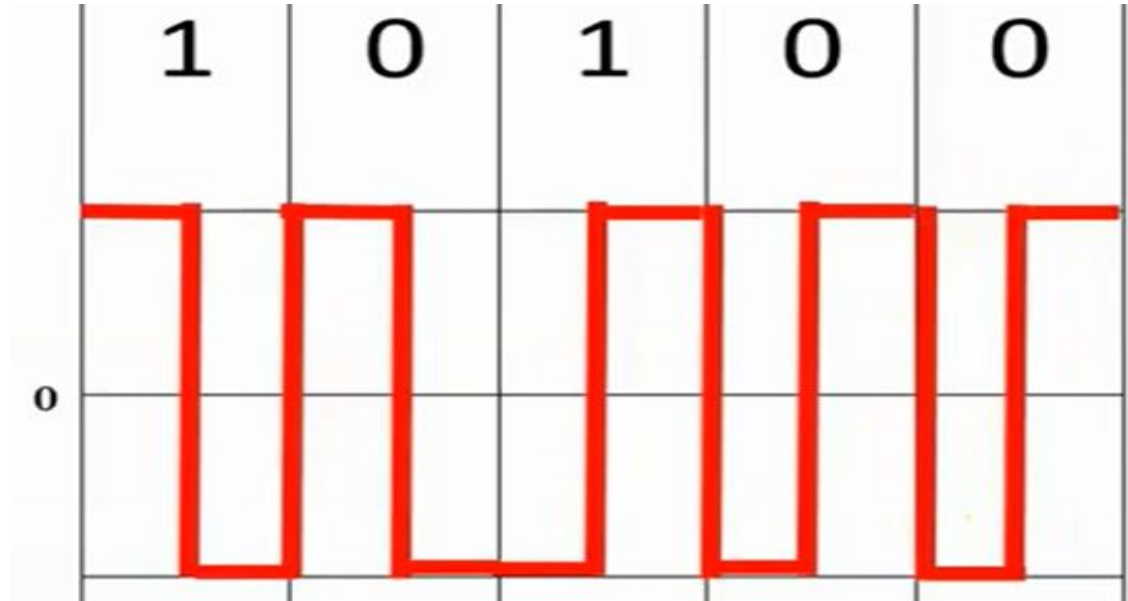


2. Polar Schemes : Differential Manchester

Combines the logic of
RZ and NRZ-I



Example 1:
Data : 1 0 1 0 0

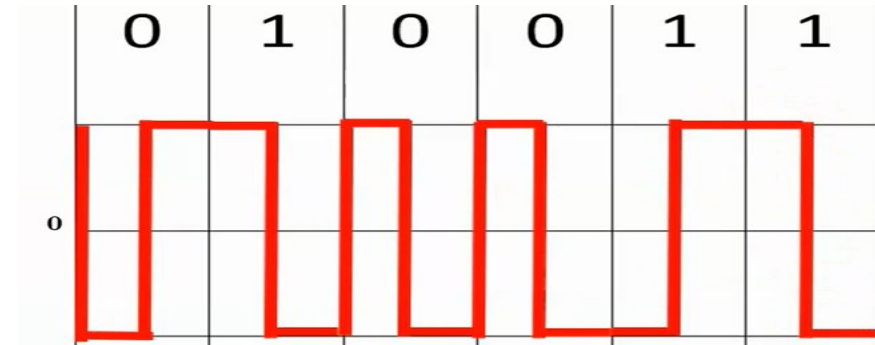
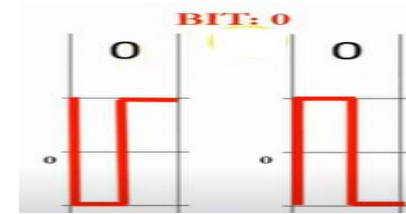
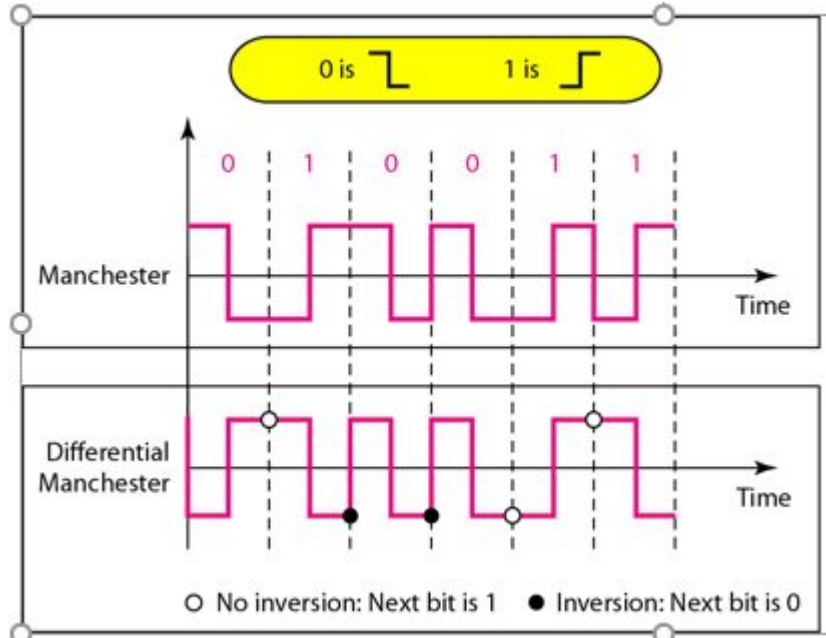


2. Polar Schemes : Differential Manchester

cont..

Example 2:

Data : 0 1 0 0 1 1

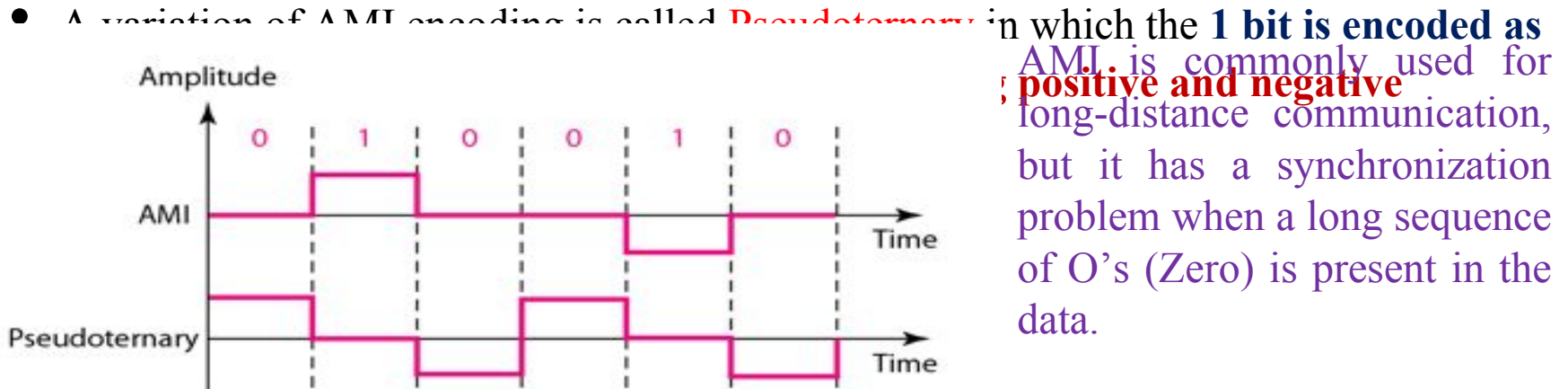


2. Polar Schemes : Differential Manchester

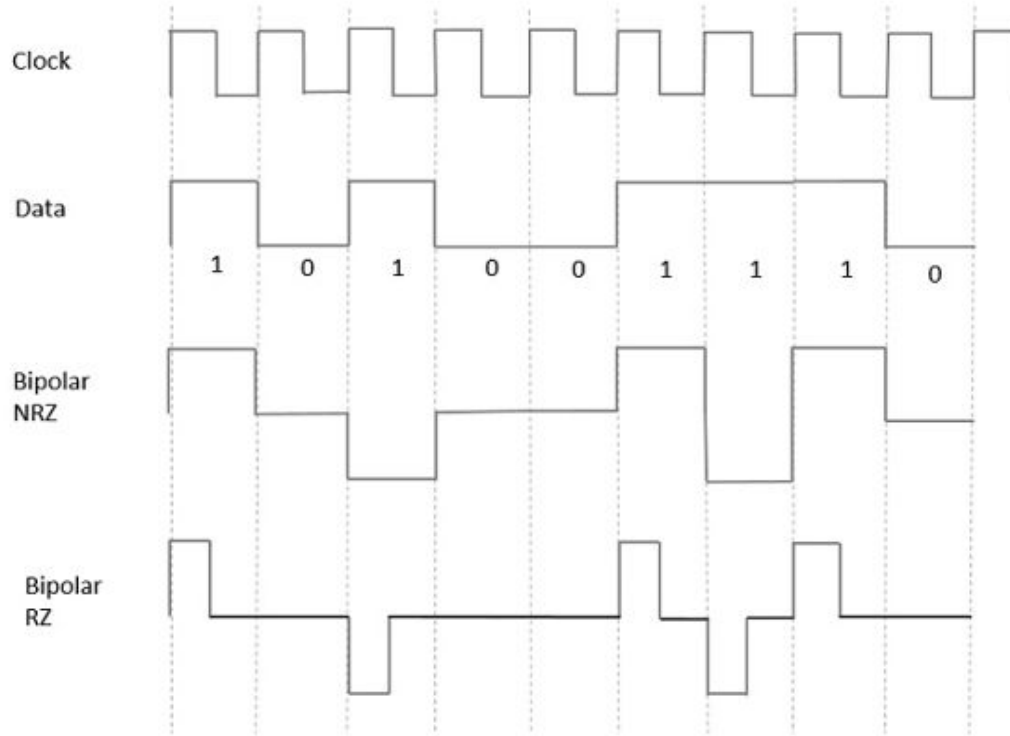
- In Manchester and differential Manchester encoding, the transition at the middle of the bit is used for synchronization
- The minimum bandwidth of Manchester and differential Manchester is 2 times that of NRZ.

3. Bipolar: Alternate Mark Inversion (AMI) and Pseudoternary (only for student Information)

- Alternate Mark Inversion (AMI) and Pseudoternary.
- **Mark** means **1**. So **AMI means alternate 1 inversion**.
- A neutral zero voltage represents binary 0. Binary 1s are represented by alternating positive and negative voltages.



Bipolar



Advantages

- It is simple.
- No low-frequency components are present.
- Occupies low bandwidth than unipolar and polar NRZ schemes.
- This technique is suitable for transmission over AC coupled lines, as signal drooping doesn't occur here.
- A single error detection capability is present in this.

Disadvantages

- No clock is present.
- Long strings of data causes loss of synchronization.

2. Block Coding

- We need **redundancy** (extra bits) bits are required during transmission: to ensure **synchronization** and to provide some kind of **inherent error detecting**.
- Block coding can give us this **redundancy and improve the performance** of line coding.
- Block coding is also called as **mB/nB** coding (m Bits, n Bits).
- Changing Block of **m Bits** to Block of **n Bits**. Where $n > m$.
A technique of sending data in **a set of sequence**.

Block coding is done in **three** steps:

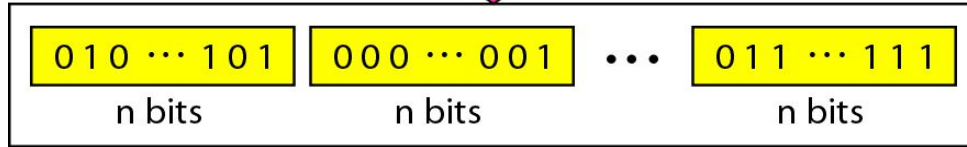
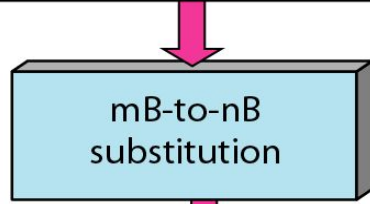
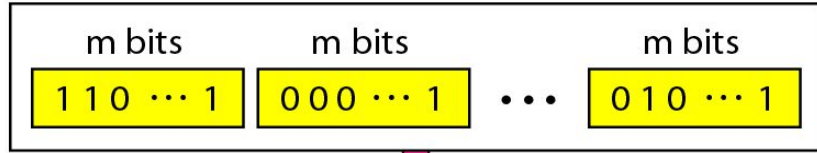
1. Division
2. Substitution
3. Combination.

Need to **add redundant** bits for padding, error checking

https://www.youtube.com/watch?v=l-C_st2Oq0A

Block Coding

Division of a stream into m -bit groups



Combining n -bit groups into a stream

1. Division

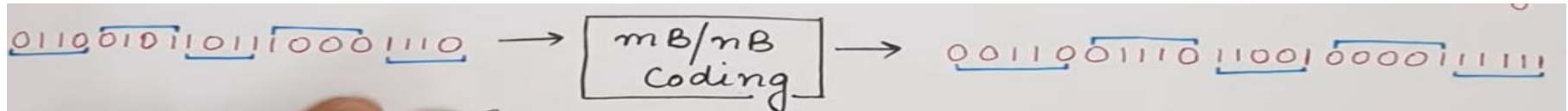
divide a sequence of bits into groups of m bits each)

2. Substitution

(replace m -bit group with an n -bit group.)

3. Combination

(n -bit groups are combined together to form a stream and this stream has more bits of data than original bits)



Analog-to-Digital Conversion

Two techniques

- 1. Pulse Code Modulation (PCM)**
- 2. Delta Modulation (DM)**

Analog-to-Digital Conversion

- It is more **robust** to noise and can easily be recovered, corrected and amplified.
- **Analog-to-Digital Conversion:** Two techniques
 1. **Pulse Code Modulation (PCM)**
 2. **Delta Modulation (DM)**

Components of PCM encoder

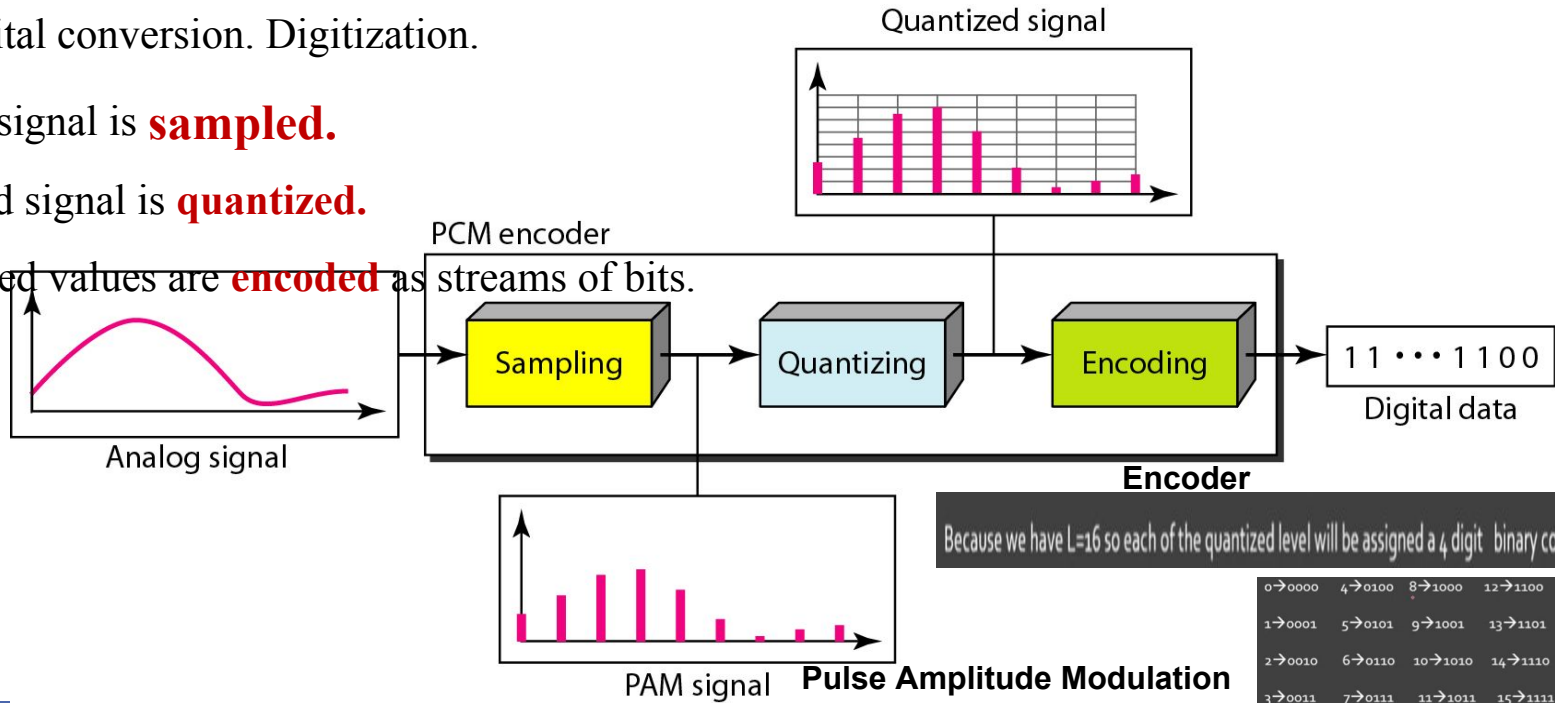
- Sampling converts a continuous time signal to discrete time signal.

Rounding off the values to one of the closest permissible numbers (or quantized levels)

Pulse Code Modulation.

Analog to digital conversion. Digitization.

1. The analog signal is **sampled**.
2. The sampled signal is **quantized**.
3. The quantized values are **encoded** as streams of bits.



Because we have $L=16$ so each of the quantized level will be assigned a 4 digit binary code

Advantages

- ❖ Uniform transmission quality
- ❖ Compatibility of different class of traffic in network
- ❖ Integrated Digital Network
- ❖ Increased utilization of existing circuit.
- ❖ Good performance over poor transmission path.

Disadvantages

- ❖ Large Bandwidth requires for transmission
- ❖ Noise and Crosstalk leaves low but attenuation will increase

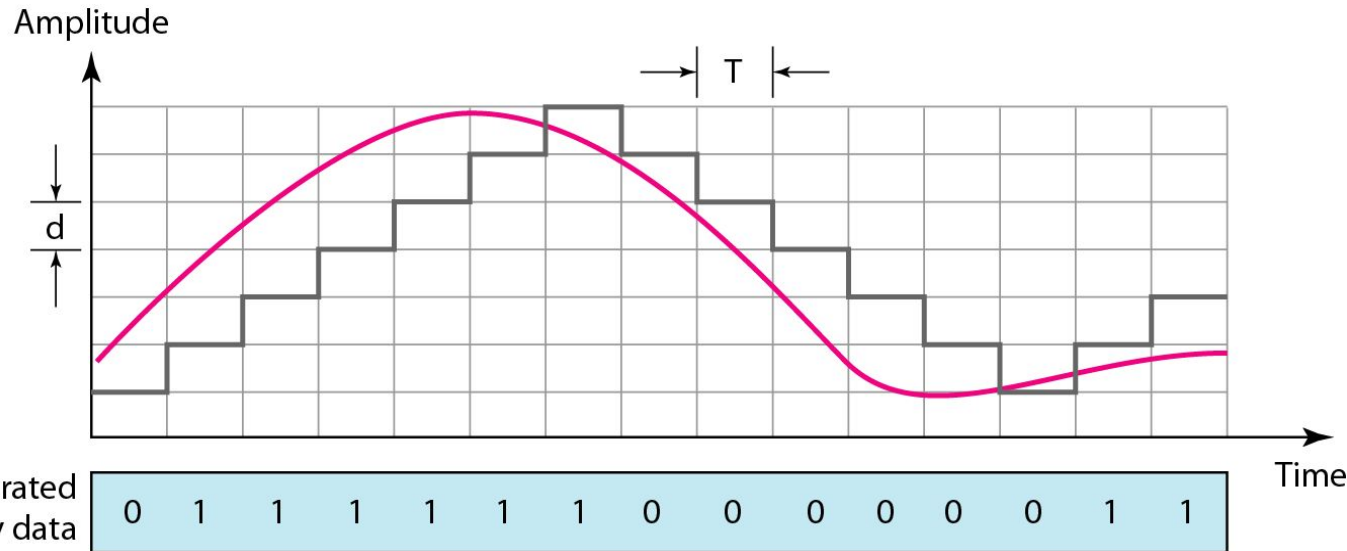
Applications

- ❖ In Compact disk
- ❖ Digital Telephony
- ❖ Digital Audio applications

Delta Modulation

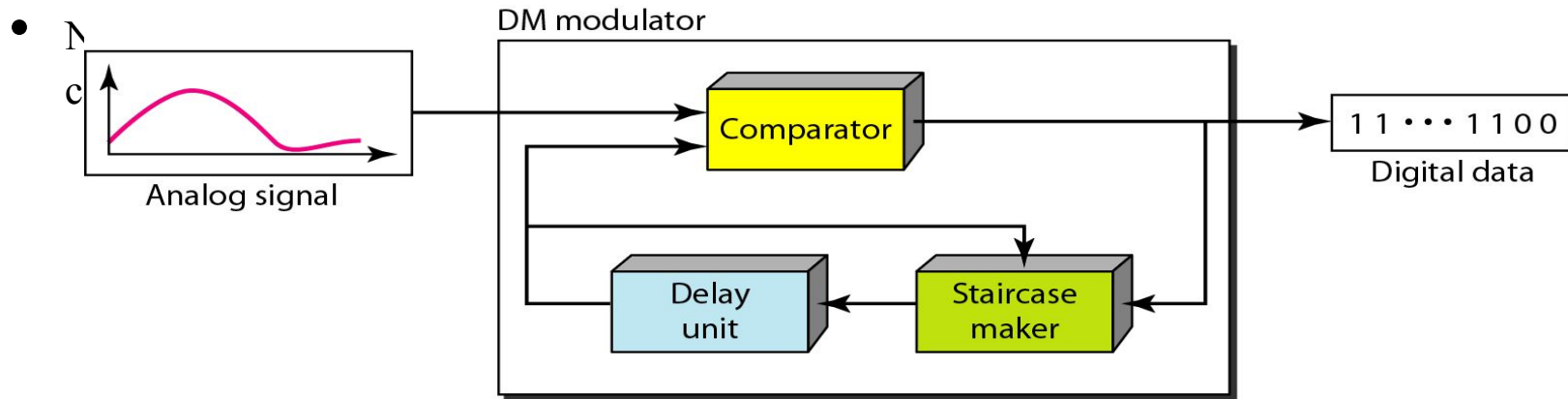
- PCM finds the value of the signal amplitude for each sample; DM finds the change from the previous sample.
- There are no code words here; bits are sent one after another.

Figure The process of delta modulation



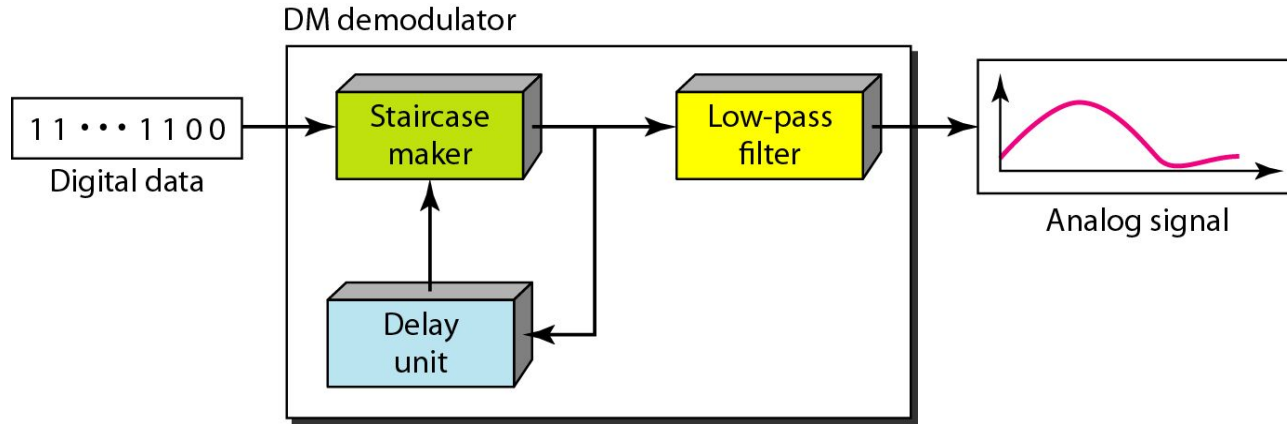
Delta Modulation Components

- Modulator: is used at the sender site to create a stream of bits from an analog signal.
- If the delta is positive, the process records a 1; if it is negative, the process records a 0.
- Base of comparison is required. Which is done by Staircase Maker.
- The modulator, at each sampling interval, compares the value of the analog signal with the last value of the staircase signal.



Delta Demodulation Components

- **Demodulator:** The demodulator takes the digital data and, using the staircase maker and the delay unit, creates the analog signal.
- Low-pass filter is used for smoothing.



Delta Demodulation Components

- **Adaptive DM:**
 - A better performance can be achieved if the value of δ is not fixed.
 - In adaptive delta modulation, the value of δ changes according to the amplitude of the analog signal.
- **Quantization Error:**
 - DM is not perfect.
 - Quantization error is always introduced in the process.
 - The quantization error of DM, however, is much less than that for PCM.

Introduction to Computer Networks

- ❑ A **computer network** is a collection of two or more devices, such as computers, servers, printers, and mobile devices, connected to exchange data and resources(files, applications, and internet access).
- ❑ Networks enable communication through wired connections (Ethernet, optical fiber) or wireless connections (Wi-Fi, Bluetooth) and follow predefined rules called **protocols** to ensure accurate and secure data transmission .

Key features

- ☐ Data sharing
- ☐ Resource sharing
- ☐ Communication
- ☐ Centralized Management



Shared Services

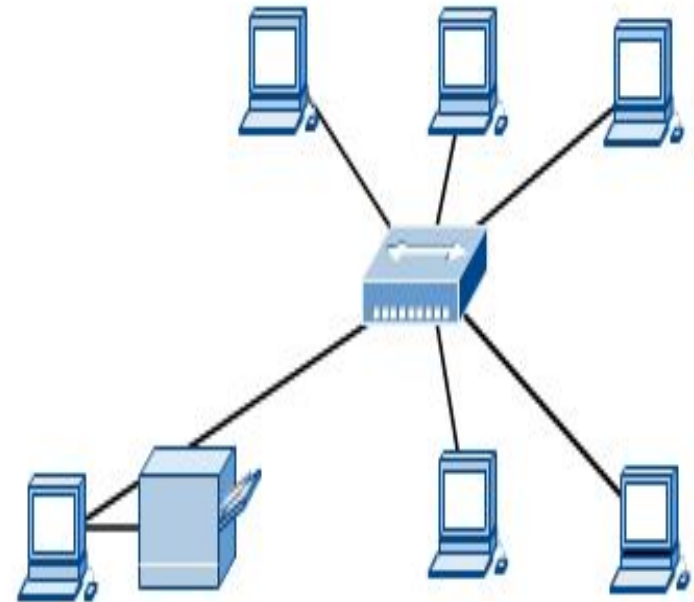
Web Service – Users can access websites or web applications hosted on a web server.

Communication Services – Services such as video conferencing, voice calls (VoIP), and instant messaging.

Authentication Service – A central server verifies user identities and manages login credentials.

Computer Networks

- A **computer network** is a group of two or more independent computers and devices connected to share data, resources, and services.
- These connections can be established using wired media or wireless
- Types of Computer Networks are classified based on factors:
 - Geographical area
 - Transmission medium
 - Ownership and access control



Types of Networks

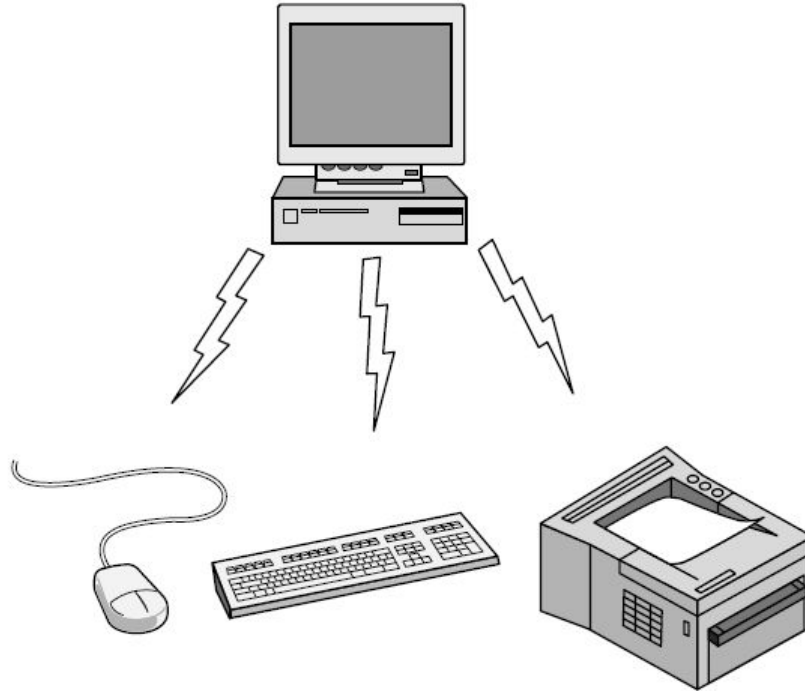
Classification of interconnected processors:

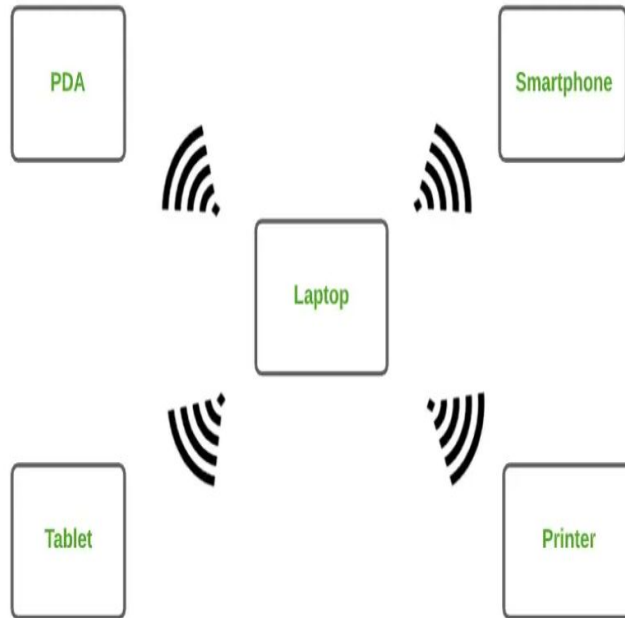
1. **Personal area networks (PAN)**
2. **Local area networks (LAN)**
3. **Metropolitan area networks (MAN)**
4. **Wide area networks (WAN)**

Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	Local area network
100 m	Building	
1 km	Campus	
10 km	City	Metropolitan area network
100 km	Country	Wide area network
1000 km	Continent	
10,000 km	Planet	The Internet

1. Personal Area Network (PAN)

Coverage Range: 1–10 meters





PAN (Personal Area Network) Working

Connects personal devices around an individual within a short range (typically up to 10 meters).

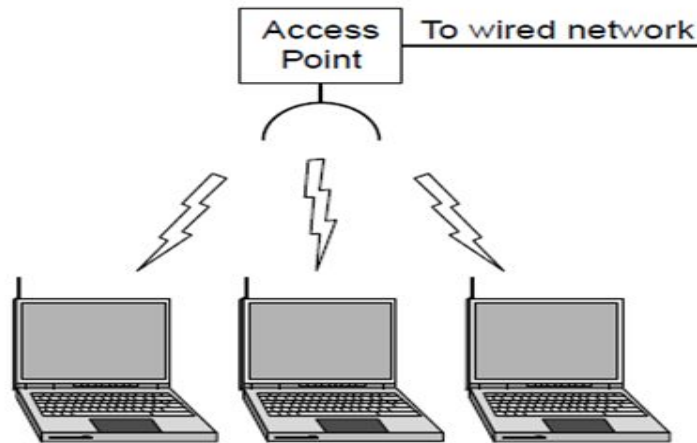


2. Local Area Network (LAN)

Coverage Range: One room, building, or small campus

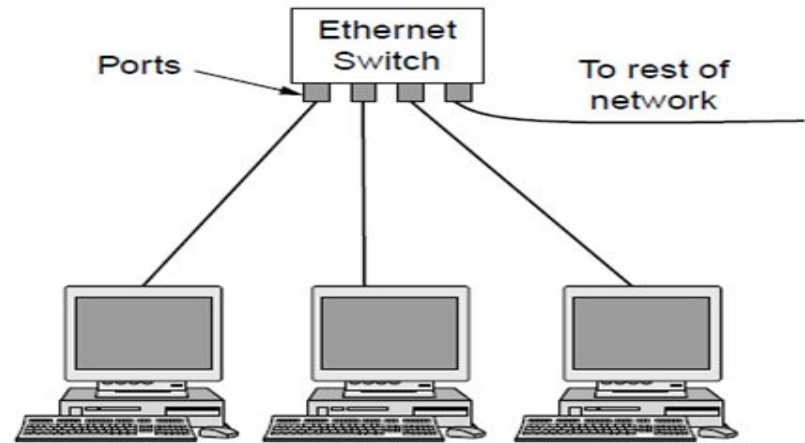
Wireless and Wired LAN

(a) 802.11



(a) 802.11 (standard which defines protocols for Wireless Local Area Networks)

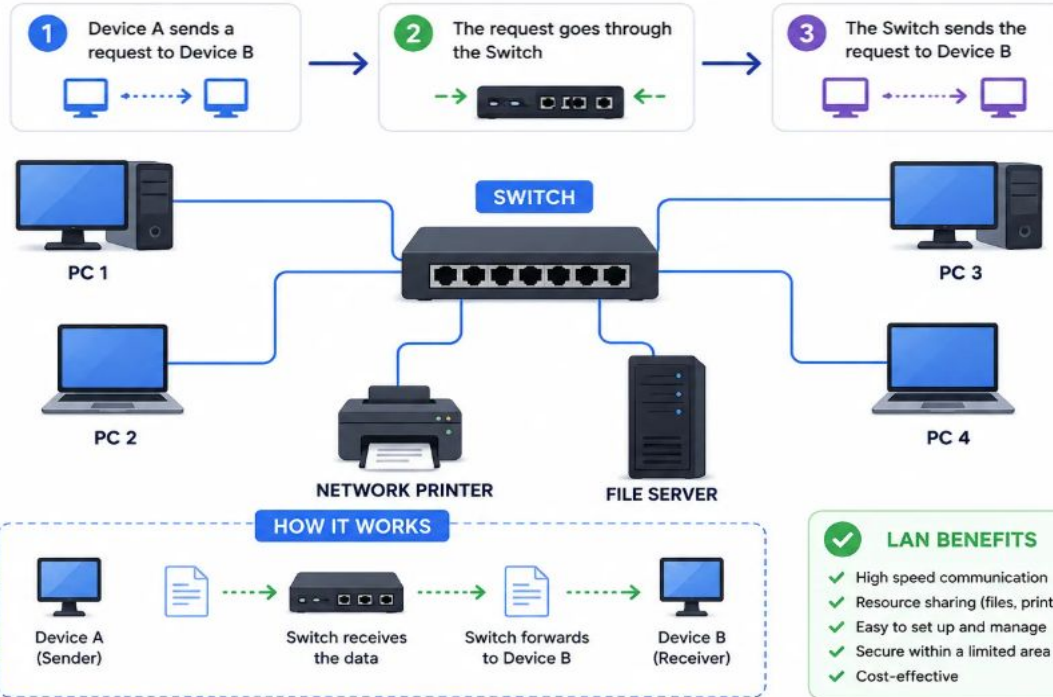
(b) Switched Ethernet



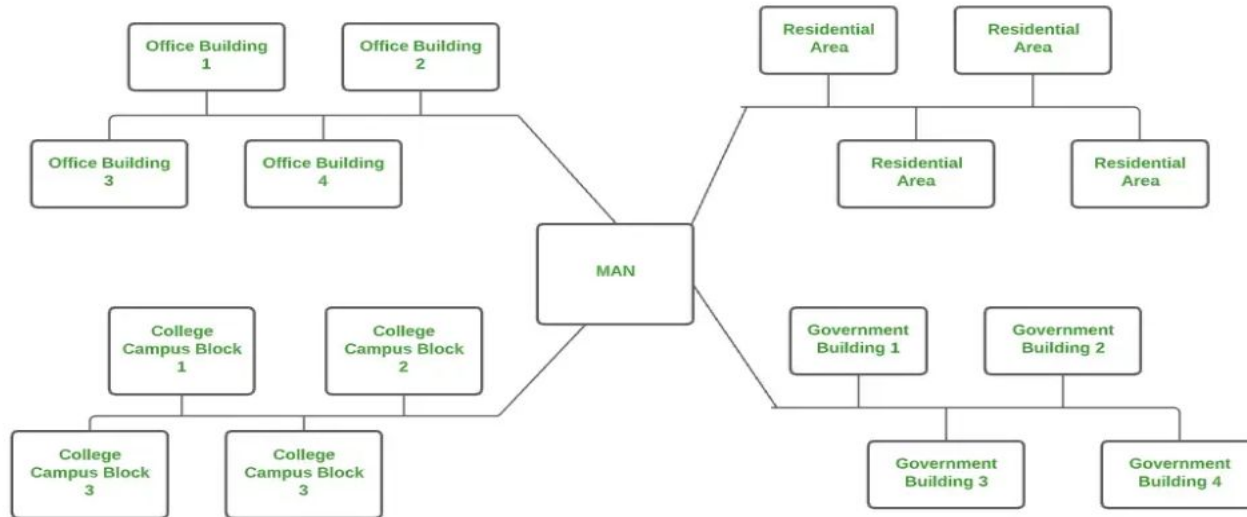
(b) Switched Ethernet (Ethernet provides services for wired media)

LAN (Local Area Network) Working

Devices within a limited area connect to each other and share resources



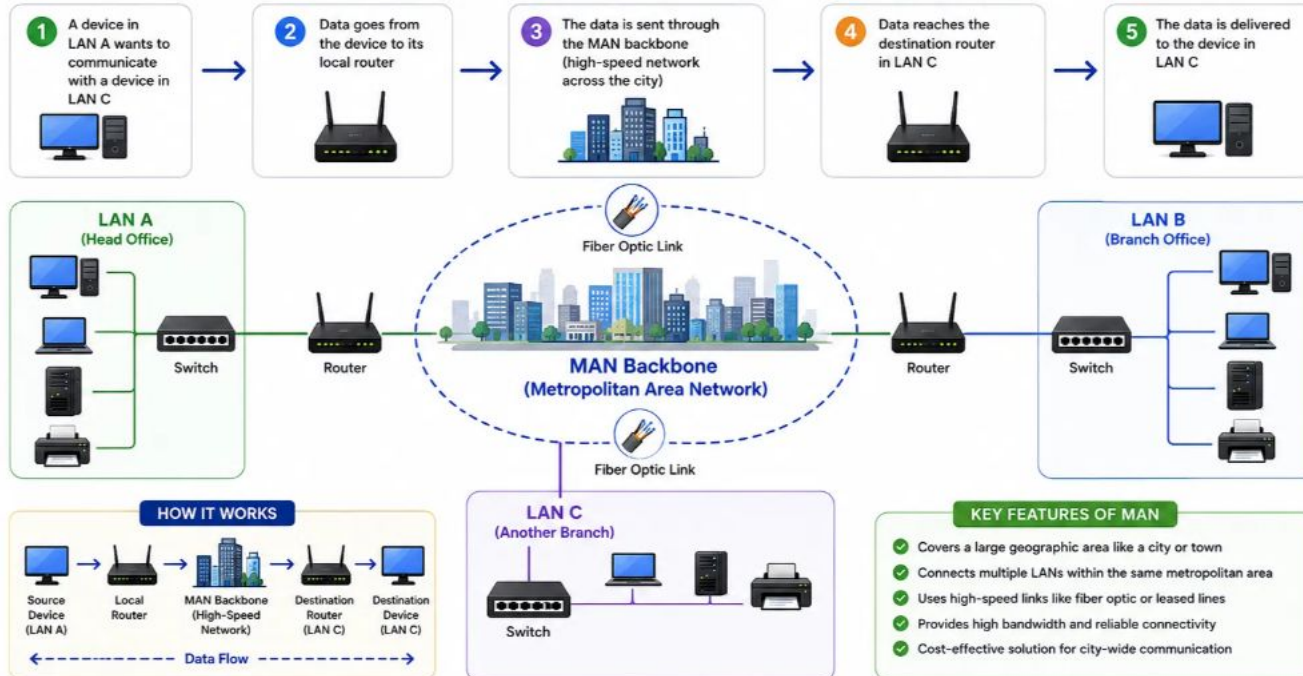
3. Metropolitan Area Network (MAN):



Metropolitan Area Network (MAN)

MAN (Metropolitan Area Network) Working

Connects multiple LANs across a city or metropolitan area using high-speed links.



EXAMPLES OF MAN



City Government Networks



University / College Campus Network



Healthcare Networks



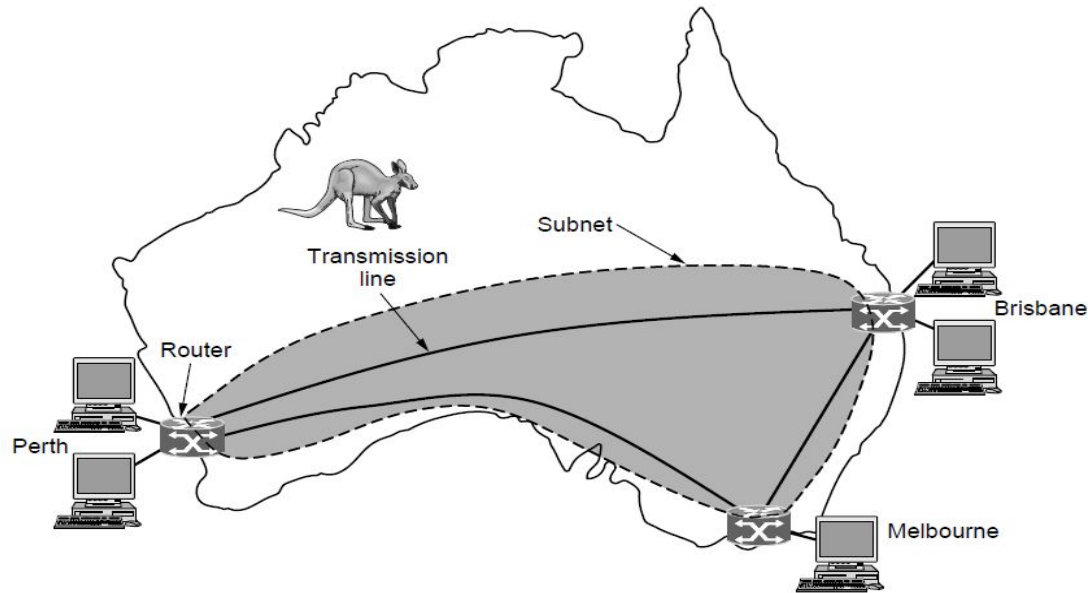
Banking & Financial Networks



Corporate Networks in a City

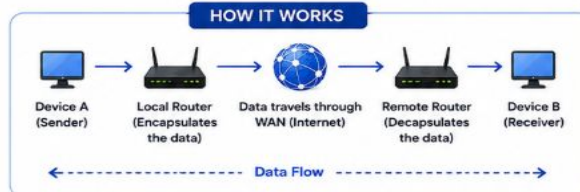
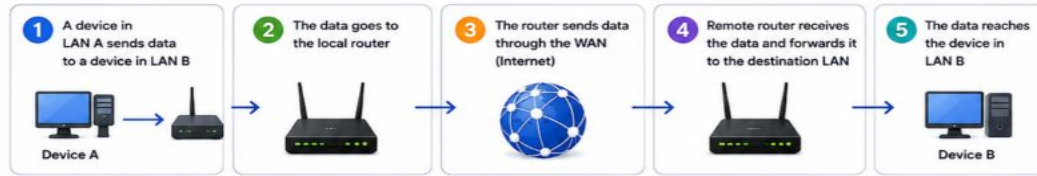
4. Wide Area Network (WAN)

WAN: connects three branch offices in Australia



WAN (Wide Area Network) Working

Connects multiple LANs across large geographical areas using routers and the Internet.



Comparison of LAN, MAN, WAN

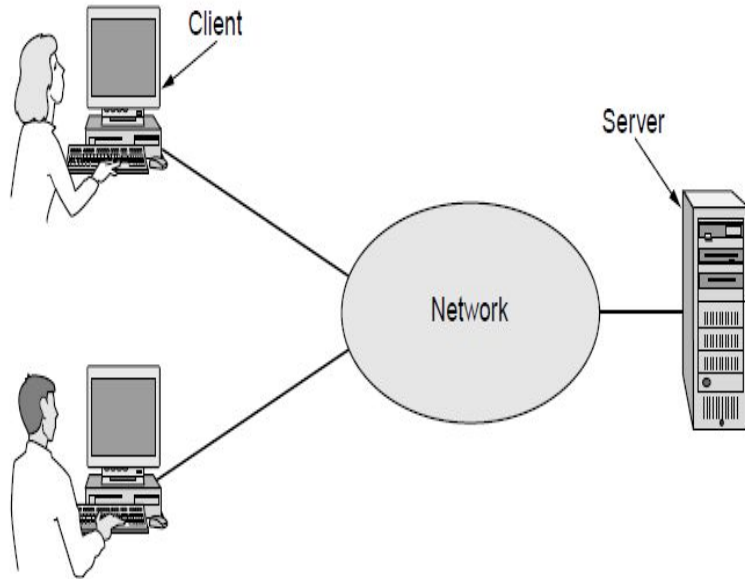
Sr.No.	Key	LAN	MAN	WAN
1	Definition	LAN stands for Local Area Network.	MAN stands for Metropolitan Area Network.	WAN stands for Wide Area Network.
2	Ownership	LAN is often owned by private organizations .	MAN ownership can be private or public .	WAN ownership can be private or public .
3	Speed	LAN speed is quite high .	MAN speed is average .	WAN speed is lower than that of LAN.
4	Delay	Network Propagation Delay is short in LAN.	Network Propagation Delay is moderate in MAN.	Network Propagation Delay is longer in WAN.
5	Congestion	LAN has low congestion as compared to WAN.	MAN has higher congestion than LAN.	WAN has higher congestion than both MAN and LAN.
6	Fault Tolerance	Fault Tolerance of LAN is higher than WAN.	Fault Tolerance of MAN is lower than LAN.	Fault Tolerance of WAN is lower than both LAN and MAN.
7	Maintenance	Designing and maintaining LAN is easy and less costly than WAN.	Designing and maintaining WAN is complex and more costly than LAN.	Designing and maintaining WAN is complex and more costly than both LAN and MAN.

Other Network Types

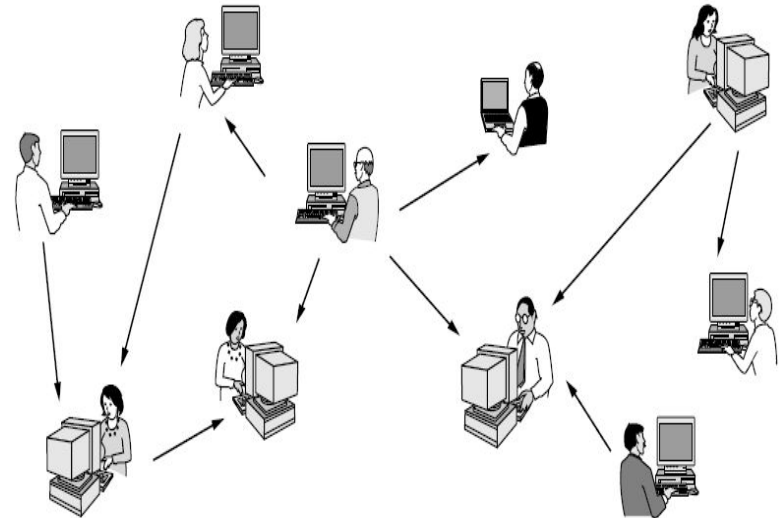
- **Wireless LAN (WLAN):** Similar to LAN but uses wireless communication like Wi-Fi.
- **Storage Area Network (SAN):** High-speed network for connecting storage devices to servers.
- **Virtual Private Network (VPN):** Extends private networks over the internet for secure remote access.

Uses of Computer Networks

Business Applications

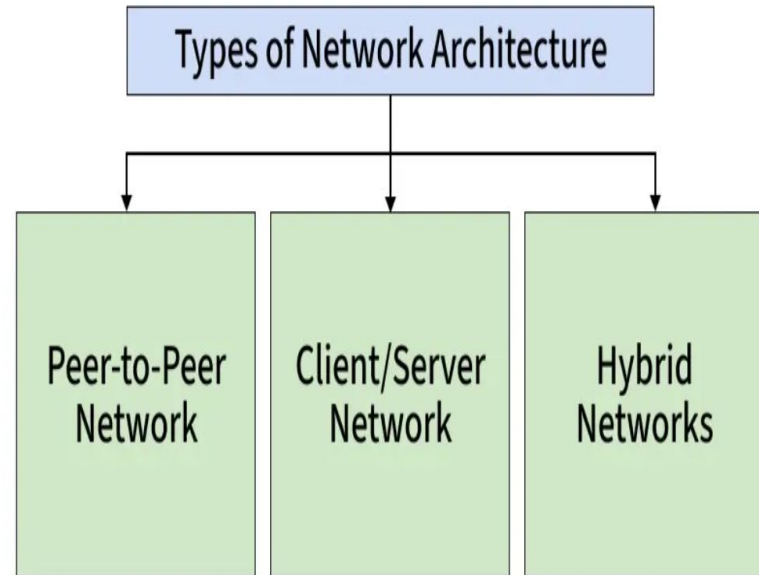


Home Applications: P2P having no fixed clients & servers



Network Architectures

1. Peer-to-Peer
2. Client-Server
3. Hybrid



Types of Network Architecture

1. Peer-to-Peer Architecture

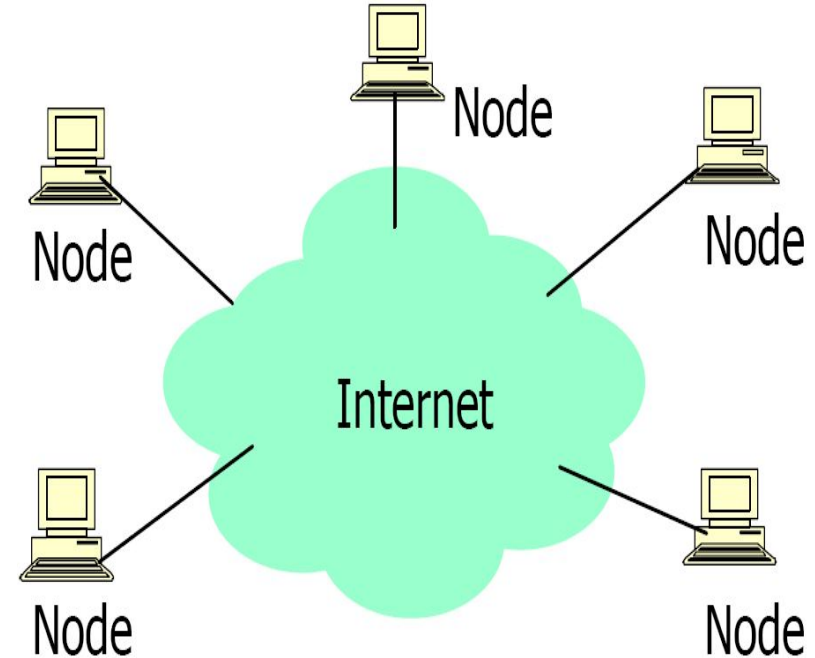


All nodes acts as a clients and servers

- Provides and consume data
- Any node can initiate a connection

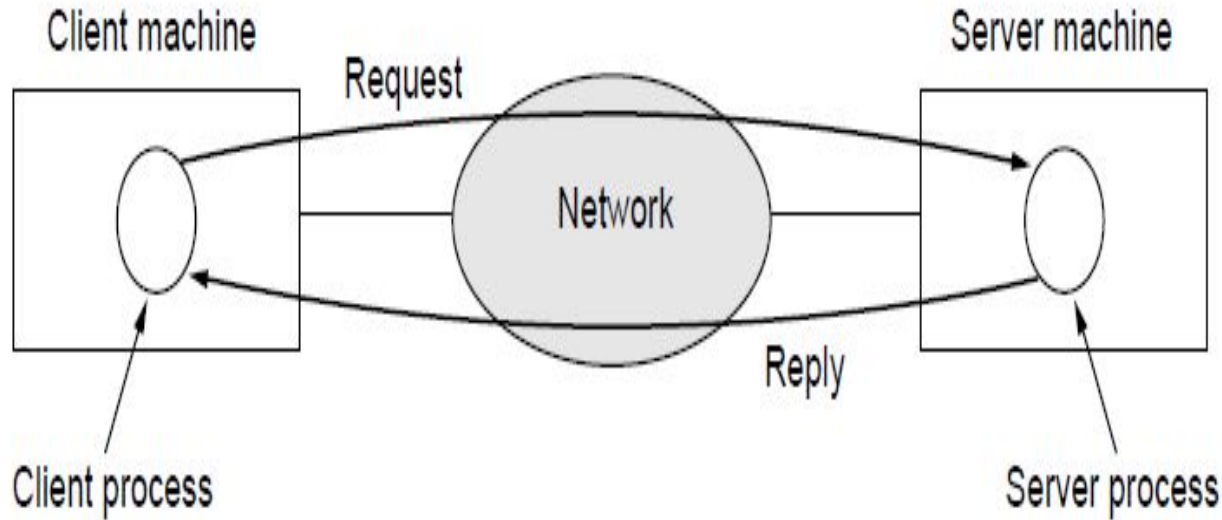


No centralized data source

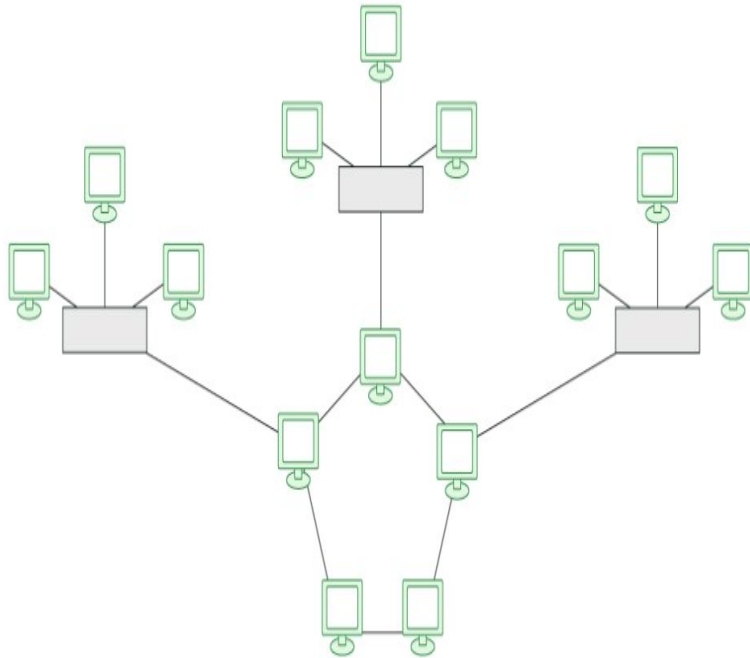


2. Client-Server Model

Client-server model: involves **Requests and Replies**



3. Hybrid Network



Hybrid Network

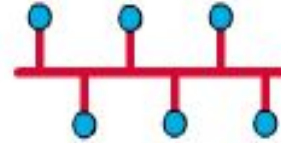
- Hybrid networks combine elements of both client-server and peer-to-peer architectures.
- They leverage the centralized control of client-server networks and the decentralized resource sharing of P2P networks.
- Hybrid networks are used in applications requiring both centralized management and distributed resource sharing.

Network Topologies:

1. Bus Topology
2. Star Topology
3. **Extended star** or Tree Topology
4. Ring Topology
5. Mesh Topology

Network Topologies

- Way of connection between computers, printers and other devices.
- Layout of the wire and devices as well as the paths used by data transmissions.



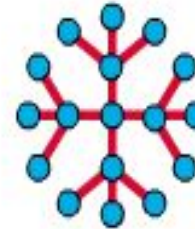
Bus Topology



Ring Topology



Star Topology



Extended Star Topology



Mesh Topology

Bus Topology

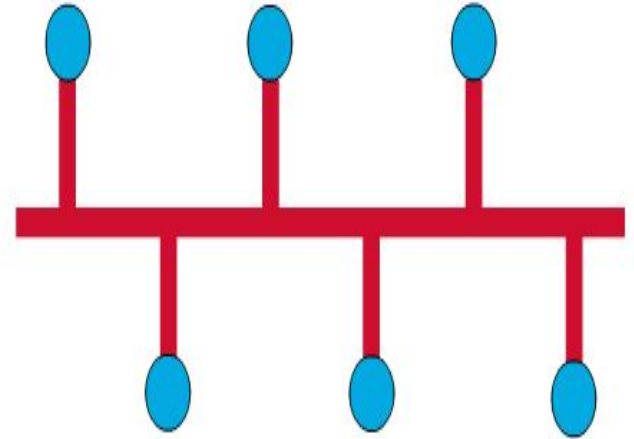
- Commonly referred as a linear bus
- All the devices are connected by one single cable

Advantages

- It is easy to set up, handle, and implement.
- It is best-suited for small networks.
- It costs very less.

Disadvantages

- The cable length is limited - limits the number of network nodes
- When the number of devices connected to the bus increases, the efficiency decreases.



Star Topology

Star topology

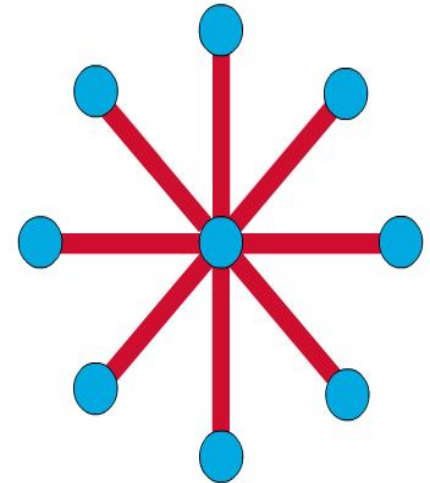
- most commonly used architecture in Ethernet LANs
- resembles spokes in a bicycle wheel

Advantages of Star Topology

- Due to its centralized nature, the topology offers simplicity of operation.
- It also achieves isolation of each device in the network.
- Adding or removing network nodes is easy.
- Due to the centralized nature, it is easy to detect faults in the network devices.

Disadvantages of Star Topology

- Network operation depends on the functioning of the central hub. Hence, central hub failure leads to failure of the entire network.
- Also, the number of nodes that can be added, depends on the capacity of the central hub.



Tree Topology

Extended star topology or tree topology

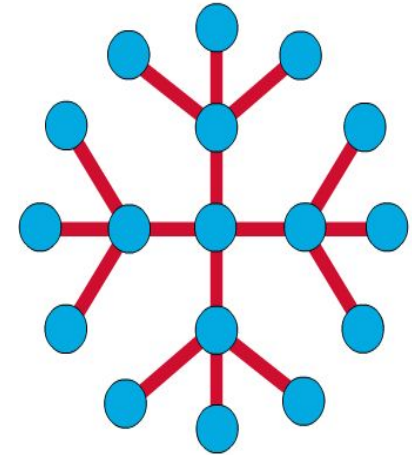
- significantly reduces the traffic on the wires by sending packets only to the wires of the destination host
- network devices that filter frames or packets like bridges, switches and routers are used .

Advantages of Tree Topology

- The tree topology is useful in cases where a star or bus cannot be implemented individually.
- Fault identification is easy.
- The network can be expanded by the addition of secondary nodes. Thus, scalability is achieved.

Disadvantages of Tree Topology

- As multiple segments are connected to a central bus, the network depends heavily on the bus. Its failure affects the entire network
- Owing to its size and complexity, maintenance is not easy and costs are high
- Though it is scalable, the number of nodes that can be added depends on the capacity of the central bus and on the cable type

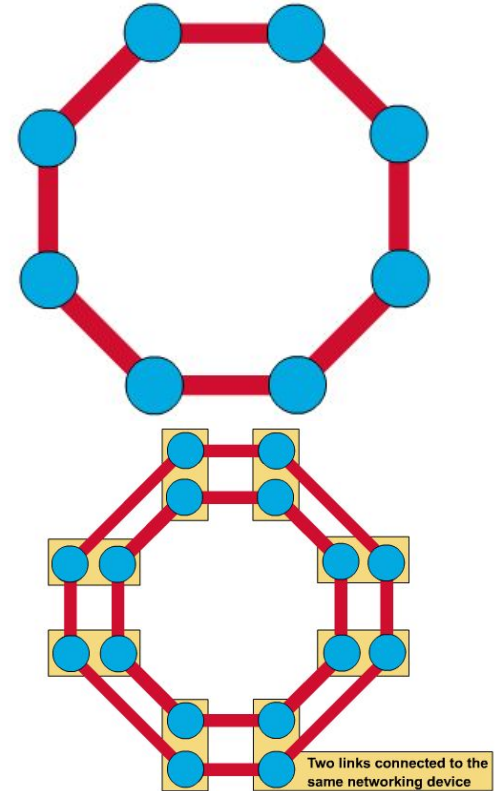


Ring Topology

- A frame travels around the ring, stopping at each node
- A Node wants to transmit data it adds the data as well as the destination address to the frame, which takes the data out of the frame
- Single ring – all the devices on the network share a single cable
- Dual ring – allows data to be sent in both directions

Advantages of Ring Topology

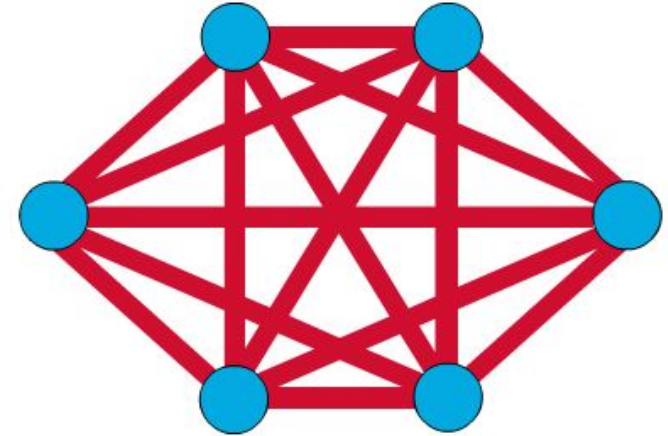
- A central server is not required
- The traffic is unidirectional and the data transmission is high-speed
- In comparison to a bus, a ring is better at handling load.
- The adding or removing of network nodes is easy.
- It is less costly than a star topology.



Disadvantages of Ring Topology

Mesh Topology

- Connects all devices (nodes) to each other for redundancy and fault tolerance
- Used in WANs to interconnect LANs and for mission critical networks like those used by banks and financial institutions
- Implementation is Expensive and difficult
- **Advantages of Mesh Topology**
- Possible to transmit data from one node to many other nodes at the same time.
- The failure of a single node does not cause the entire network to fail
- It can handle heavy traffic



- **Disadvantages of Mesh Topology**
- A lot of cabling is required.
- Costly setup and maintenance
- Administration of a mesh network is difficult

Comparison of network Topologies

Topology	Information Transfer	Setup	Expansion	Troubleshooting	Cost	Cabling Concerns
Star Each computer connects to a central connection device.	All information passes through the central network connection.	Each computer must be close to the central device. 100 meters maximum cable length. Up to 24 computers per network.	Add a new computer by plugging in a new cable from the computer to the connection device.	When one computer goes down, the rest of the network is unaffected. If the connection device goes down, then the network is down.	More expensive of the simple topologies, it requires costly connection device. Usually cheaper than a hybrid network.	Uses twisted pair cable. Requires large amounts of cable. No more than 100 meters from the computer to the connection device.
Bus Single cable connects everything.	One computer at a time sends information. Information goes along the cable and the computer accesses the information off the cable.	Connect the cable from one computer to the next and so on to the end. A terminator is placed at each end of the network.	To add a computer, you must shut down the network and disconnect the cable from the existing computers.	If one computer malfunctions, the entire network goes down.	A cheaper network since there is usually one continuous copper cable.	Single continuous cable connects the devices. Terminator is required at each end of the cable. Uses coaxial or twisted pair cabling.

Comparison of network Topologies

cont-

Ring Single cable configured in a ring.	Information goes in one direction around the ring and passes along the ring until it reaches the correct computer.	Computers are located close to each other. Setup is easy. There is no connector. The ring has no beginning and no end.	Cable between the computers must be broken to add a new computer, so the network is down until the new device is back online.	If there's a break in the cable or an error in the network, information continues to transfer through the rest of the ring until reaching the point of the break. This makes troubleshooting easy.	One of the more expensive topologies due to high cable costs.	Requires more cabling than other topologies. Uses twisted pair.
Hybrid Mesh Combines two or more different structures.	Often used across long distances. Information transfer can happen in different ways, depending on the other topologies.	Often created when expanding an existing network. Can use a variety of connection devices.	Connection devices make combining different networks and different topologies easy.	Troubleshooting is most difficult in this topology because of the variety of technologies.	Expensive, large, and usually complicated.	Cabling depends on the types of networks. Can use twisted pair and coaxial cable. Also incorporates fiber optic cabling over long distances.

Transmission Media

Transmission Media

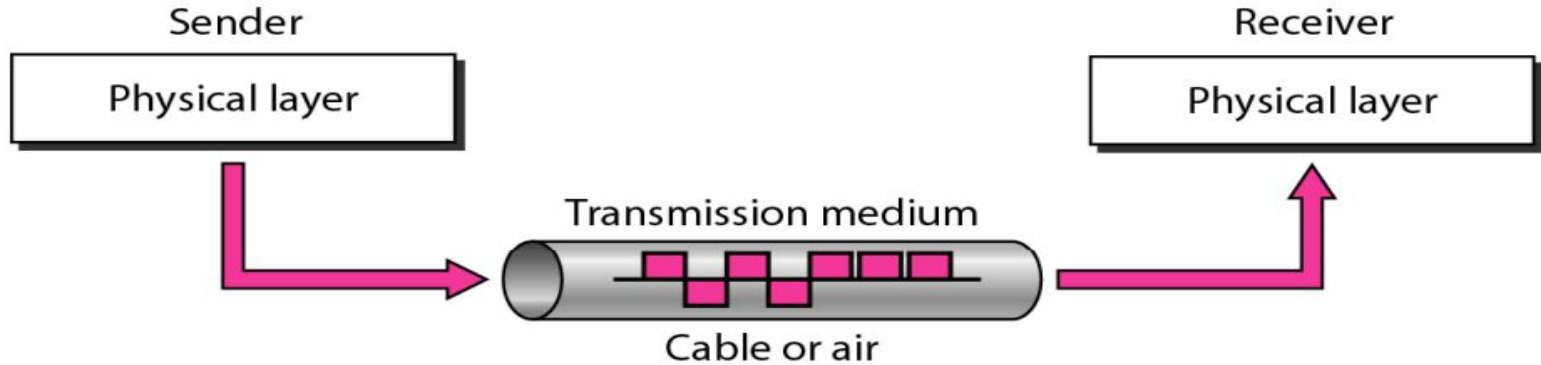
□ Transmission media is a **communication channel** that carries the information from the **sender** to the **receiver**.

□ **Properties:**

bandwidth,

delay,

cost and ease of installation and maintenance.



Transmission medium and physical layer

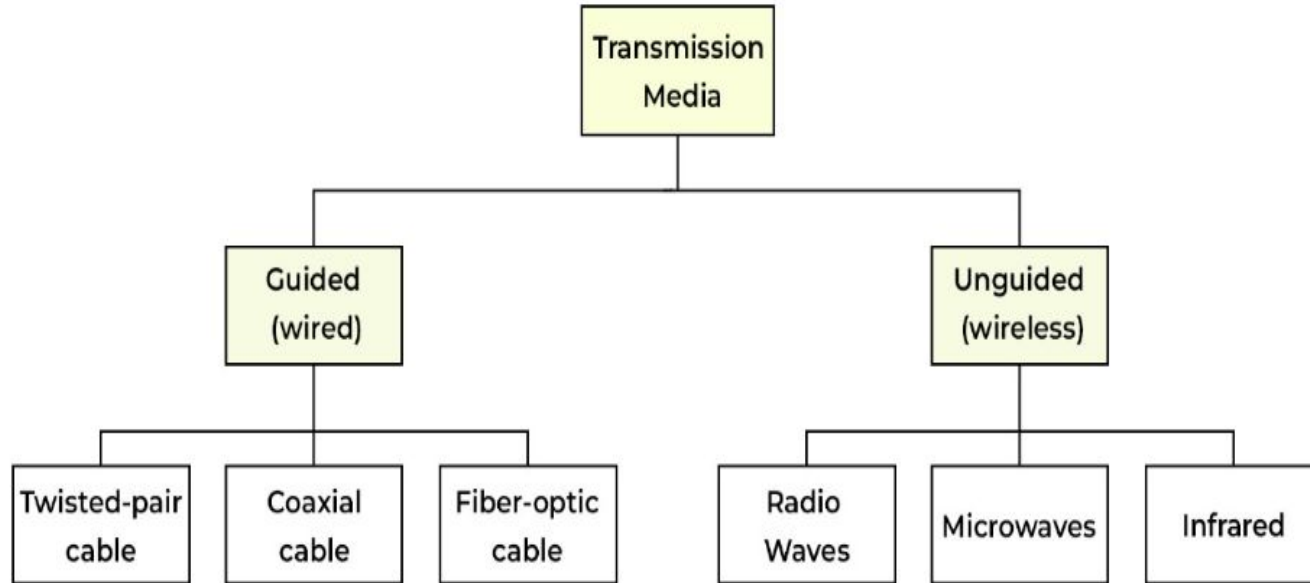
Transmission Media

Media Terminology

1. **Segment length** – The signal degrades for each type of media after traveling a certain distance to the point where it becomes unintelligible. This distance is referred to as the segment length.
2. **Attenuation** – The loss of signal strength when traveling a distance is called attenuation. Measured in decibels (DB).
3. **Bandwidth** – Amount of data that can travel through the cable in a unit period of time. Measure in kbps, Mbps, etc.
4. **Interference** – Each cable is susceptible to certain forms of external noise. This is called EMI (Electromagnetic Interference) or RFI (Radio frequency interference)
5. **Crosstalk** – When a signal carrying conductor is placed near another signal carrying conductor. Interference is generated in the other cable. This is called crosstalk.

Classification of Transmission Media

(depending on the nature and quality of transmission)



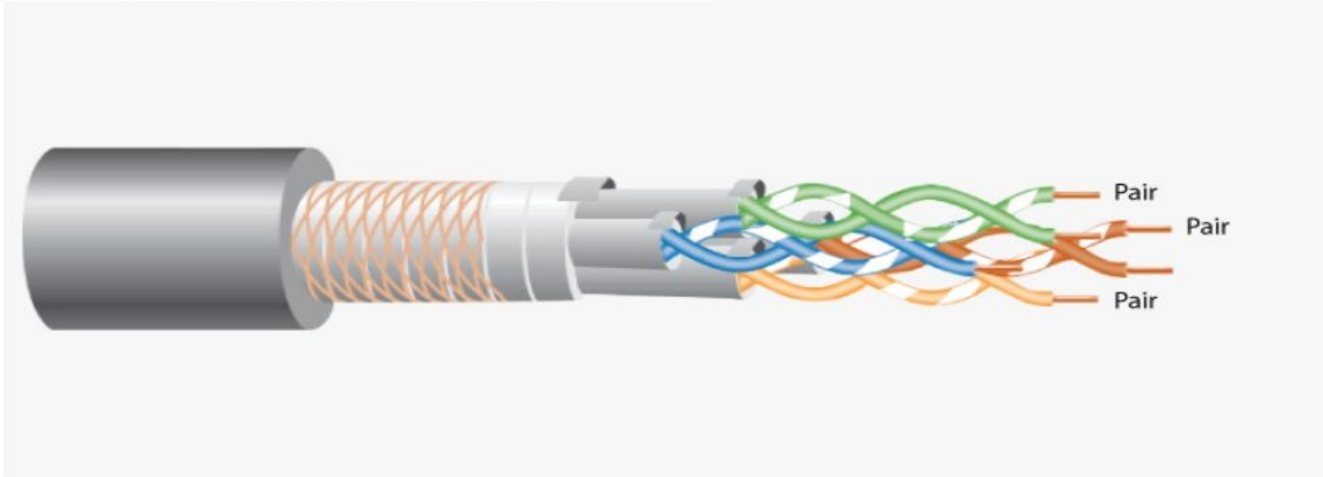
Twisted-Pair Cable

- This cable has eight insulated wires.
- These are paired in groups of 2 and are twisted together based on a color code.
- The twisting is done to decrease interference caused by the adjacent wires.
- One wire in the pair may send signals to the receiver, while the other serves as a ground reference.



a. Shielded twisted pair

These twisted pair cables are covered in a braided shield which acts as a shield from outside interference.



Advantages of Shielded Twisted Pair

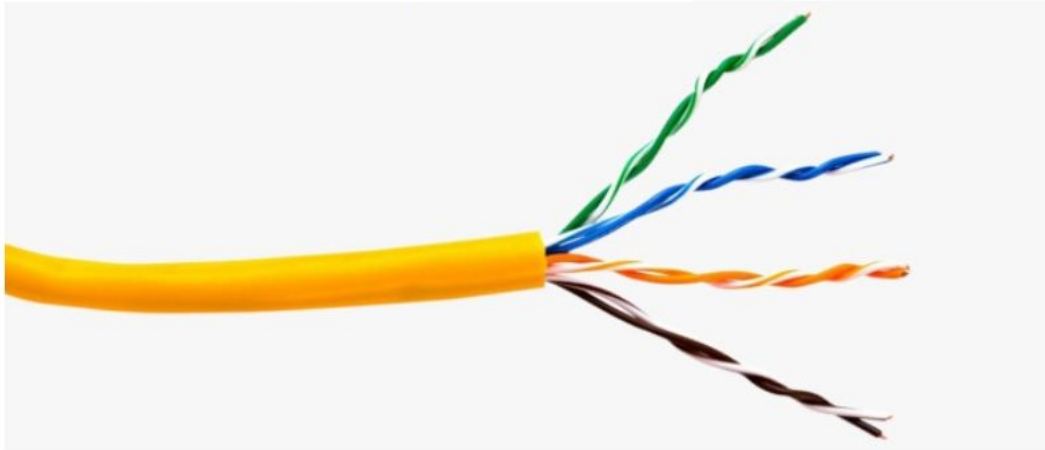
- Shielded Twisted Pair is capable of eliminating crosstalk.
- It is faster compared to Unshielded Twisted pair.

Disadvantages of Shielded Twisted Pair

- It is more expensive.
- It is more bulky.
- It is difficult to manufacture and install.

b. Unshielded twisted pair

These twisted pair cables do not have a braided shield. The 4 pairs are simply covered in a plastic insulator for safety.

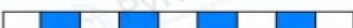


Color Codes –

TIA/EIA 568A Wiring

1		White and Green
2		Green
3		White and Orange
4		Blue
5		White and Blue
6		Orange
7		White and Brown
8		Brown

TIA/EIA 568B Wiring

1		White and Orange
2		Orange
3		White and Green
4		Blue
5		White and Blue
6		Green
7		White and Brown
8		Brown

Straight through cable

- When the same color codes are used at both ends. For ex, 568A on both ends. Alternatively, 568B on both ends
- Used when connecting **different types of devices**, for ex, connecting a PC to a switch, or a **switch** to a **router**

Crossover cable

- When different color codes are used at both ends. For ex, one end is 568A, and the other end is 568B
- Used when connecting similar types of devices. For ex, A PC to a PC or a switch to a switch, etc.

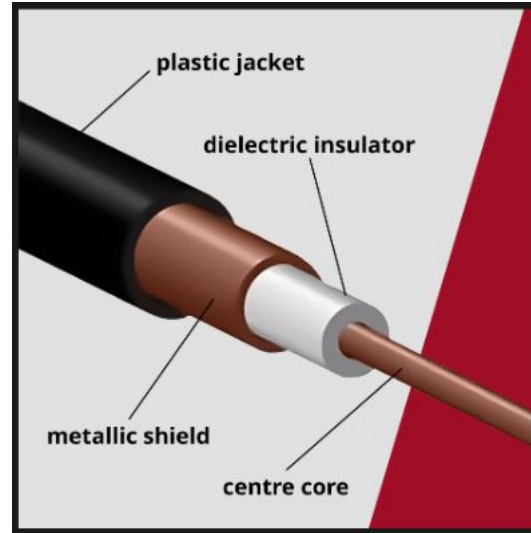
Advantages of Unshielded Twisted Pair

- Less expensive.
- It is easy to install.

Disadvantages of Unshielded Twisted Pair

- Lower capacity.
- Not suitable for long distance transmission.

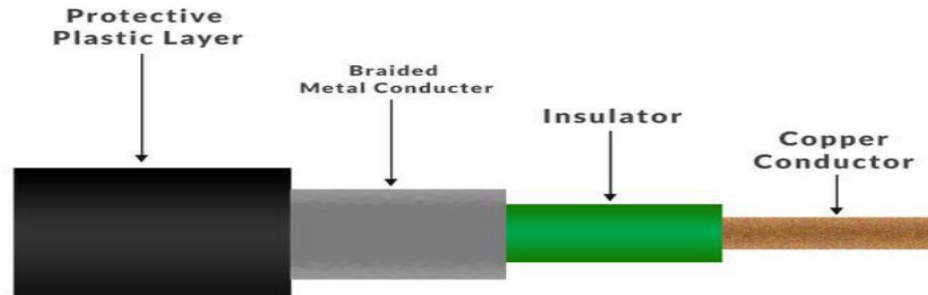
Coaxial Cable



Applications

- widely used in analog telephone networks
- Cable TV networks
- Traditional Ethernet LANs

- The core is made up of copper conductors. Its purpose is the signal transmission.
- To prevent the copper conductor from overheating, an insulator is utilized.
- A metal conductor is braided around the insulator.
- It aids in blocking out any unwanted noise or allowing for any unwanted cross-talk between electrical signals.
- The setup is entirely covered in a protective plastic layer.



Advantages of Coaxial Cable

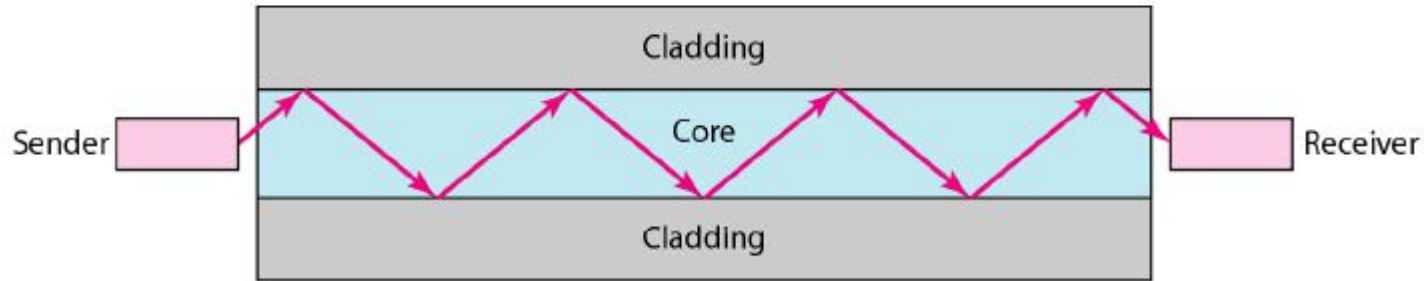
- It is easier to install.
- It has better shielding.
- It is capable of transmitting data over longer distances.
- Coaxial cables are less affected by noise or cross-talk.

Disadvantages of Coaxial Cable

- Coaxial cables are more expensive than others.
- They are less flexible and bulkier.
- They must be grounded in order to avoid crosstalk.

Fiber-Optic Cable

- ❑ Fiber optic cable is a cable that uses **light signals** for **communication**.
- ❑ It holds the optical fibers coated in plastic that are used to **send the data by pulses of light**.
- ❑ The plastic coating protects the optical fibers from heat, cold, electromagnetic interference from other types of wiring.
- ❑ It provides **faster data transmission** than copper wires.



Advantages of Fiber Optics Cable

Here are some advantages of fibre optics:

- Extremely High-speed data transmission.
- It is capable of long distance transmission without signal disruption.
- It is immune to electromagnetic intervention.

Disadvantages of Fiber Optics Cable

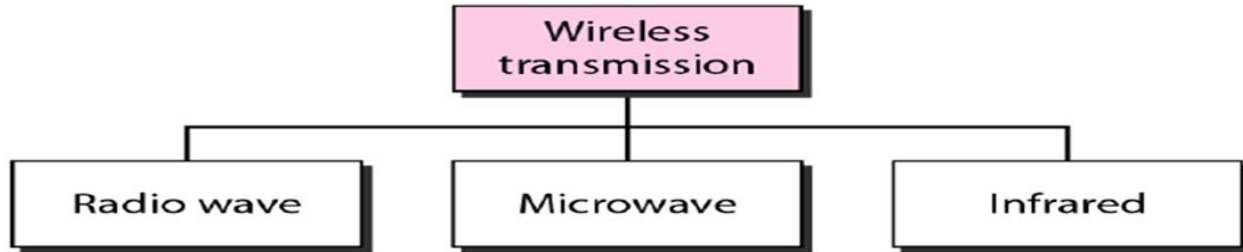
Here are some disadvantages:

- It is very expensive to install and maintain.
- It is very fragile in nature and need special handling.

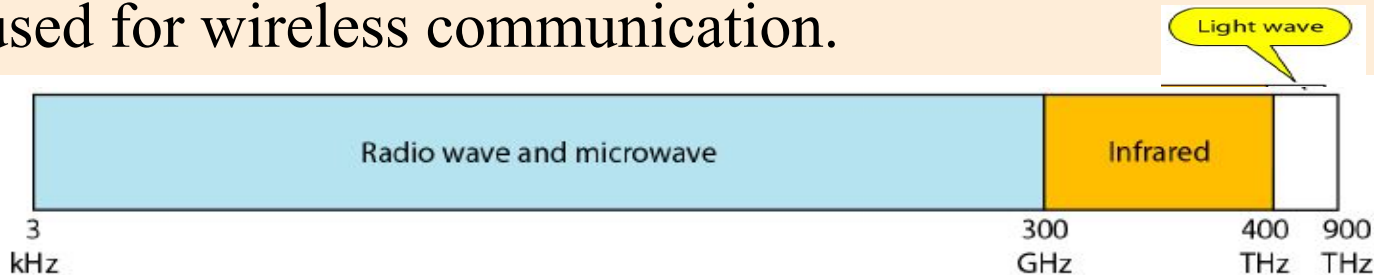
That's all from Wired Transmission Media, let's move on to Wireless Transmission media.

UNGUIDED MEDIA: WIRELESS

- An unguided transmission transmits the **electromagnetic waves without** using any physical medium. Therefore it is also known as wireless transmission.



- The electromagnetic spectrum, ranges from **3 kHz to 900 THz**, used for wireless communication.



UNGUIDED MEDIA: WIRELESS

Radio Waves

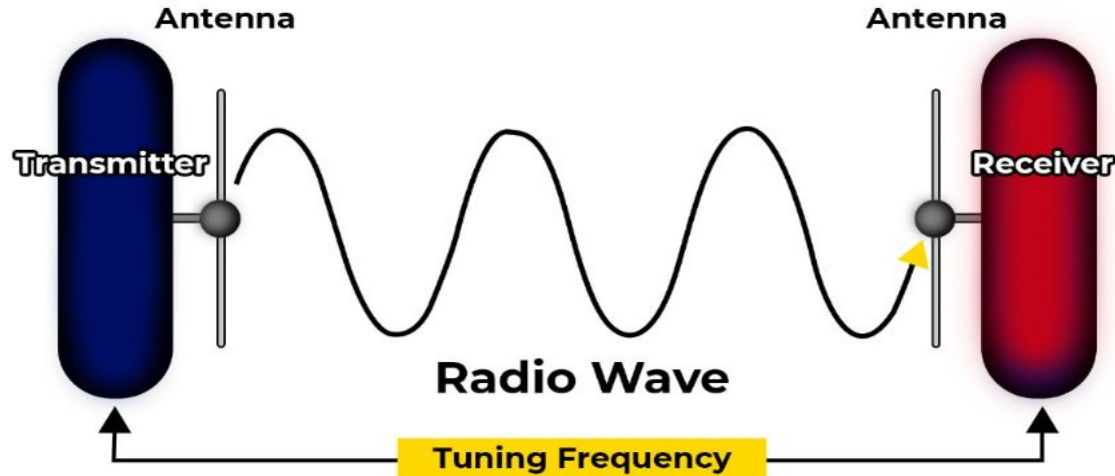
- **Electromagnetic waves** ranging in frequencies between 3KHz to 1 GHz are called radio waves

Direction of Communication

Radio waves can be directional, which means that the waves are focused in a specific direction. Also, radio waves can be omnidirectional, i.e., propagated in all directions.

Role of Antenna

An antenna is a crucial component of radio wave transmission, which is responsible for converting electrical energy into radio waves.



Application

- Radio broadcasting
- Mobile communication
- Wireless networking
- Radar and navigation

Advantages of Radio Waves

- Long-distance Communication
- Portable
- Reliable Communication
- Easy Installation

Disadvantages of Radio Waves

- Prone to Interference
- Atmospheric Disturbances
- Limited Bandwidth
- Health Risks

UNGUIDED MEDIA: WIRELESS

Microwaves

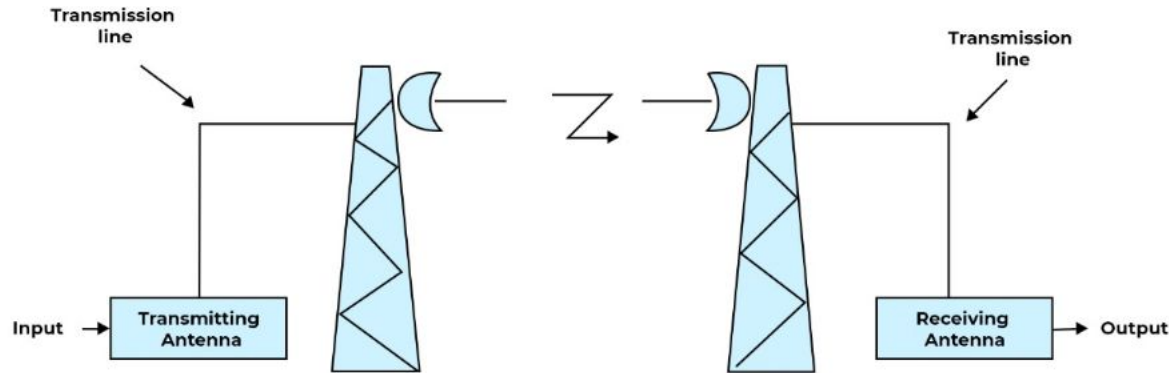
- **Electromagnetic waves** having frequencies between **1GHz and 300 GHz** are called microwaves.

Direction of Communication

- Microwaves are line-of-sight (LOS) communications. It simply means that the transmitting and receiving antennas must be in direct sight of each other.
- Microwaves are Unidirectional.

Role of Antenna

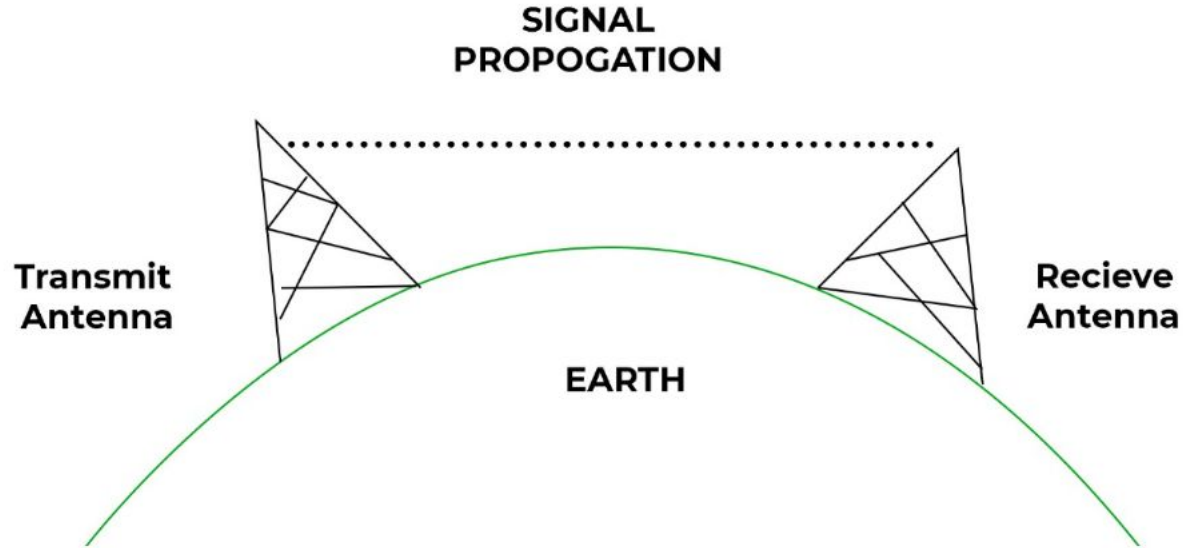
The antenna plays a crucial role in microwave transmission, as it converts electrical signals into microwave energy and transmits them through the air.



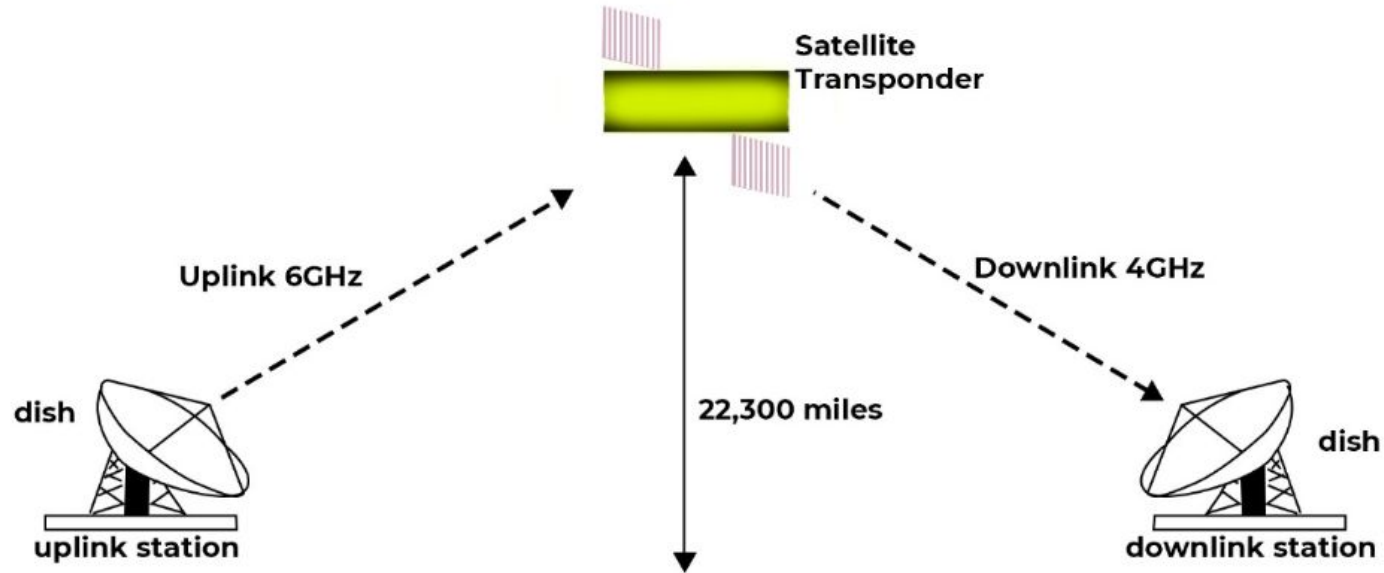
Types of Microwave Transmission

There are two types of microwave transmission. These are:

- **Terrestrial Microwaves:** These microwaves are used for communication purposes, especially between two points on the Earth's surface. One such example is the communication between two towers or buildings.



- **Satellite Microwaves:** These microwaves are used for communication between the Earth and a satellite in orbit. It is crucial for global communication and broadcasting.



Applications of Microwave Transmission

- Wireless local area networks (WLANs)
- Satellite communications
- Radar systems
- Wireless broadband internet
- Point-to-point communication links
- Radio astronomy
- Military communications
- Weather radar systems

Advantages of Microwave Transmission

- High-speed data transfer
- Easy to install and set up
- Cost-effective solutions
- Reliable communication with minimal downtime
- Fast deployment

Disadvantages of Microwave Transmission

- Interrupted by physical obstacles
- Obstructions in the line of sight can affect signal quality
- Security risks
- Limited range
- Atmospheric conditions can impact microwave signal quality

Infrared

- Frequencies from **300 GHz to 400 THz**
- Used for **short-range communication**

How it works?

- In order to send information with Infrared waves, we need special devices known as transceivers.
- These devices can send as well as receive infrared light.
- For infrared communication to work, it is recommended that the two devices that need to communicate with one another should be in sight of each other.
- In simple words, they need to be facing each other.
- If not, the light can bounce off a light-colored surface like a ceiling or a wall in order to reach the other device.

Applications

- Wireless Keyboards and Mouse
- TV Remote Control
- Night Vision
- Weapon System

Advantages of Infrared

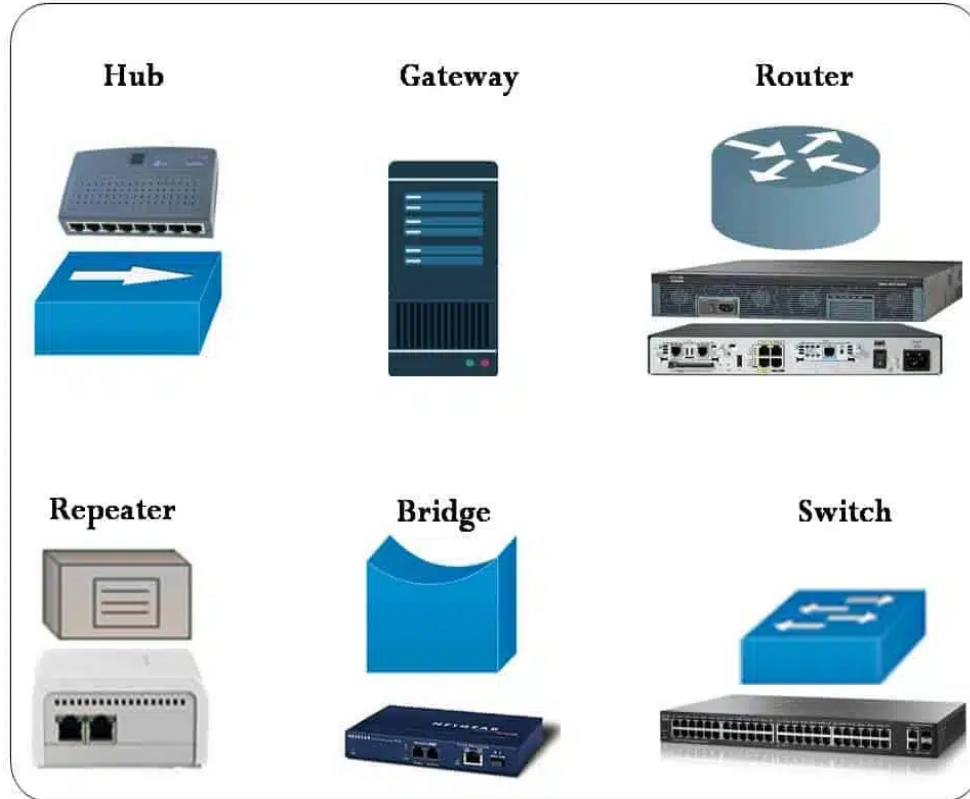
- Secure and high-speed data transfer
- Low Power Consumption
- Relatively directional
- Easy to Build into Devices

Disadvantages of Infrared

- Line of Sight Requirement
- Limited Range
- High Attenuation

Transmission Media	Common Applications
Twisted Pair Cable	Ethernet LANs, offices, schools, and home networks
Coaxial Cable	Cable TV networks, broadband internet connections, CCTV systems
Fiber Optic Cable	Data centers, ISP backbone networks, cloud infrastructure, submarine communication cables
Radio Waves	Wi-Fi networks, mobile communication, Bluetooth devices
Microwave Transmission	Point-to-point communication, cellular backhaul, satellite communication
Infrared Transmission	TV remote controls, short-range device communication, infrared sensors

Networking Devices



Application layer	Application gateway
Transport layer	Transport gateway
Network layer	Router
Data link layer	Bridge, switch
Physical layer	Repeater, hub

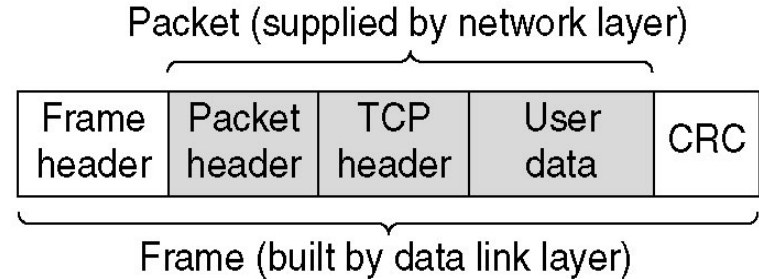
Layer wise Networking Devices

(a) Layer wise devices

Application layer	Application gateway
Transport layer	Transport gateway
Network layer	Router
Data link layer	Bridge, switch
Physical layer	Repeater, hub

(a)

(b) Frames, packets and headers



(b)

Networking Devices

Repeater

- Also known as “Signal Booster”.
- The main function of the repeater is to regenerate an incoming signal from the sender before retransmitting it to the receiver.
- It operates at the physical layer of the OSI model.



Networking Devices

Hub

- Hub is used to connect devices to the same network (**Physical layer**).
- **Hubs cannot filter data**, so data packets are sent to all connected devices



Networking Devices

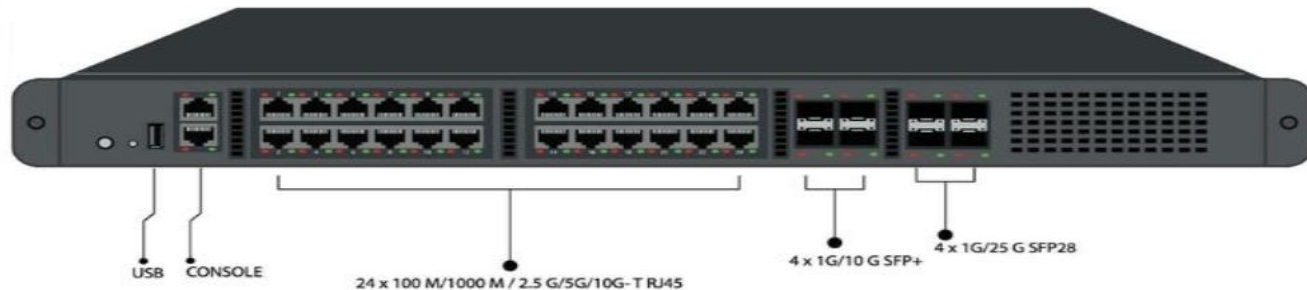
Bridge

- A bridge operates at **data link layer**
- It is also used for **interconnecting** two LANs working on the same protocol.



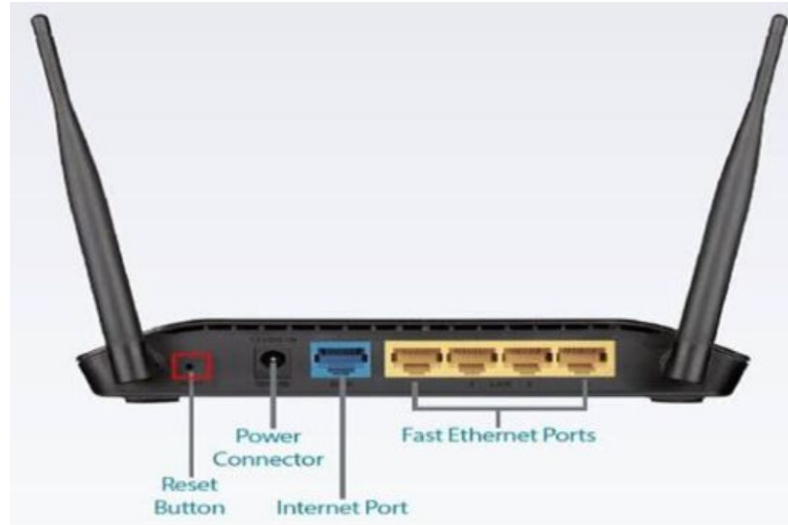
Switch

- A switch is a **data link layer device**
- It is an **intelligent network device** that can be conceived as a multiport network bridge.
- It uses **MAC addresses** to send data packets to selected destination ports
- Switches can perform some error checking before forwarding data.
- **MAC Address:** Medium access control: unique 48-bit hardware numbers of a computer that are embedded into a network card (known as a Network Interface Card) during manufacturing. The MAC Address is a unique identifier for each network interface card (NIC) on a network.



Router

- Routers operate at the **Network Layer (Layer 3)**
- Interconnect **IP** networks



Gateway:

- A gateway is simply a device or hardware that acts as a **"gate" between the networks.**
- A gateway is a network node used in telecommunications that connects two networks with different transmission protocols together
- **The gateway converts information, data or other communications from one protocol or format to another**
- An Internet gateway can transfer communications between an enterprise network and the Internet



Comparison of network devices

HUB

Hub is a broadcast device.

Hub works in the physical layer of OSI model.

Hub is used to connect devices to the same network.

Hub sends data in the form of bits.

Hub works in half duplex.

Only one device can send data at a time.

Hub does not store any MAC address of a node in the network.

SWITCH

The switch is a multicast device.

The switch works in data link layer.

The switch is used to connect devices to the network.

The switch sends data in the form of frames.

The switch works in full duplex.

Multiple devices can send data at a time.

It uses MAC addresses

ROUTER

The router is a routing device.

The router works in the network layer of OSI model.

The router is used to connect two different networks.

The router sends data in the form of packets.

The router works in full duplex.

Multiple devices can send data at a time.

It uses IP Addresses

OSI (Open Systems Interconnection) Reference Model and TCP/IP Model

Data communication is a process **or** act in which we can send or receive data.

For data communication **two models** are available: These models describe how information is transmitted between two devices across a network.

1. **OSI:** It was developed in 1984 by the International Organization for Standardization (ISO).
2. **TCP/IP:** It was designed by Vint Cerf and Bob Kahn in the 1970s.

OSI Reference Model

OSI Reference Model

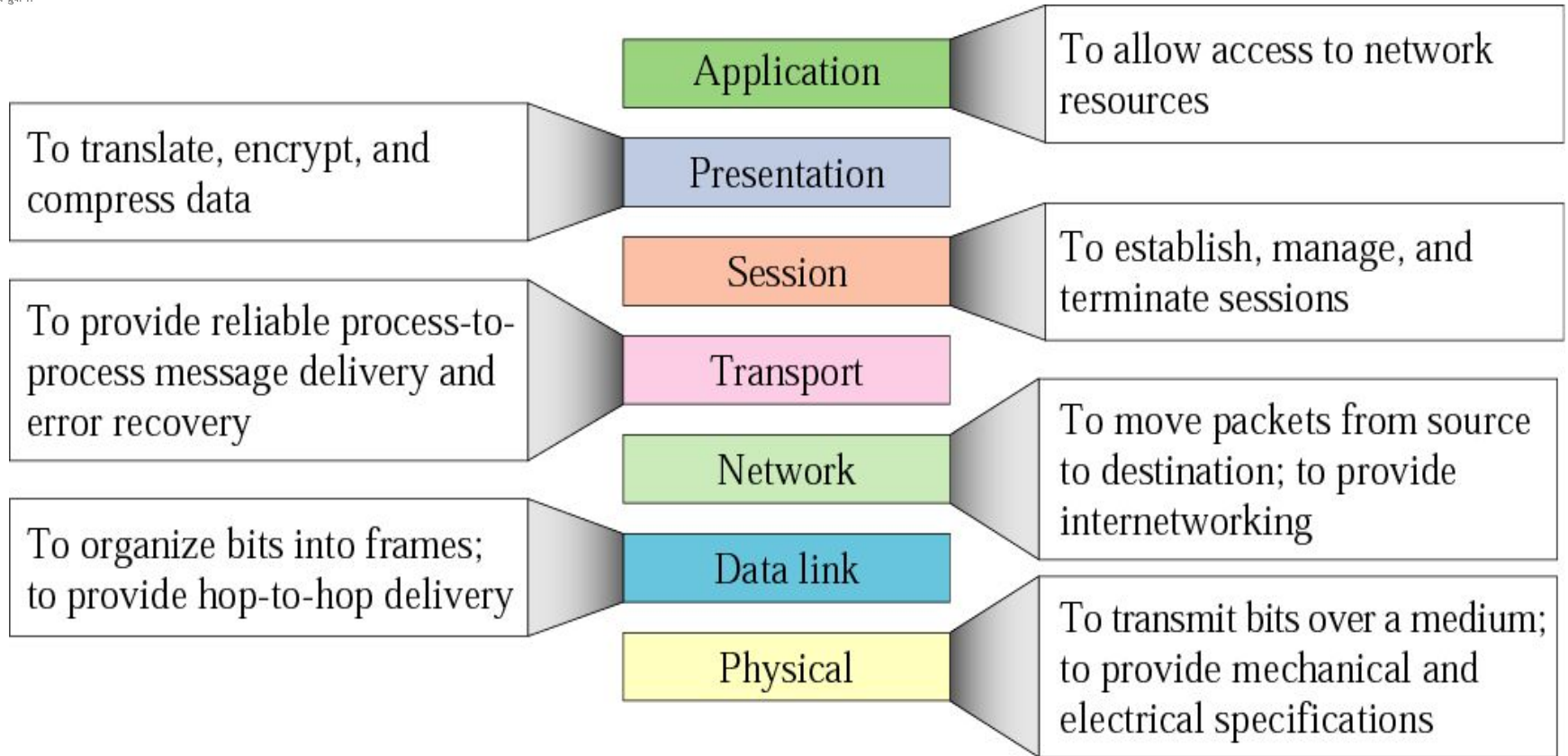
OSI: Open Systems Interconnection

The OSI provides a **standard** for different computer systems to be able to **communicate with each other**.

- Layers created for different abstractions
- Each layer performs **well-defined function**
- Function of layer chosen with definition of **international standard protocol**
- It was funded primarily by the U.S. Department of Defense.

The **Open Systems Interconnection model (OSI model)** is a [conceptual model](#) from the [International Organization for Standardization](#) (ISO) that "provides a common basis for the coordination of standards development for the purpose of systems interconnection"

OSI Reference Model: duties for each layer



1.PHYSICAL LAYER

- Bit Synchronization** Physical layer provides the bit synchronization of bits by providing a clock. This clock controls the sender and receiver providing synchronization at bit level.
- Bit Rate Control** Physical layer defines the transmission rate. The number of bits sent per second.
- Physical Topologies** The physical layer specifies how the different devices are arranged in a network (bus, ring, star, and mesh topology).
- Transmission media/mode** The physical layer checks if the transmission is simplex, half-duplex, or full-duplex. It defines how the data flows between the two connected devices and using which transmission media.

2.DATA LINK LAYER

- **Framing** Breaks messages into frames and reassembles frames into messages.
- **Error handling** It is used to soles the damaged, lost, and duplicate frames.
- **Flow Control** It keeps a fast transmitter from flooding a slow receiver.
- **Access Control** In access control, if many hosts have usage of the medium, When a single communication channel is shared by multiple devices, the MAC sub-layer of data link layer helps to determine which device has control over the channel at a given time.

3. NETWORK LAYER

- The network layer provides details that enable data to be routed between devices.
- It provides
 - the source and destination network addresses,
 - subnet information,
 - and source and destination node addresses.
- Delivering data packets from the source to the destination across multiple networks.

4.TRANSPORT LAYER

- **Segmentation and Reassembly** In this, a message is divided into small pieces. Reassemble the message correctly upon arriving at the destination.
- **Reliability** It ensures that packets arrive at their destination. Reassembles out-of-order messages.
- **Service Decisions** It is used to check what types of service to provide error-free point-to-point, datagram, etc.
- **Mapping** It determines which messages belong to which connections.
- **Naming** It must be translated into an internal address and route, send to node XYZ.
- **Flow Control** It keeps a fast transmitter from flooding a slow receiver.
- **Error Control** To retransmit the damaged segments.

5.SESSION LAYER

- Network log-on and log-off procedures
- User authentication
- Determines the type dialog available simplex, half-duplex, and full-duplex.
- Synchronization of data flow for recovery purposes.
- Creation of dialog units and activity units.

6.PRESENTATION LAYER

- The presentation layer receives any data transmitted by the application layer and prepares it for transmission over the session layer.
- It specifies how the end-user applications should format the data.
- This layer provides for the translation between the local representation of data and the representation of data that will be used for transfer between the end-users.
- The result of encryption, data compression, and virtual terminals are examples of translation services.

7.APPLICATION LAYER

- This layer provides distributed information services and controls the sequence of activities within an application and
- Controls the sequence of events between the computer application and the user of the application.
- It communicates directly with the users application program.

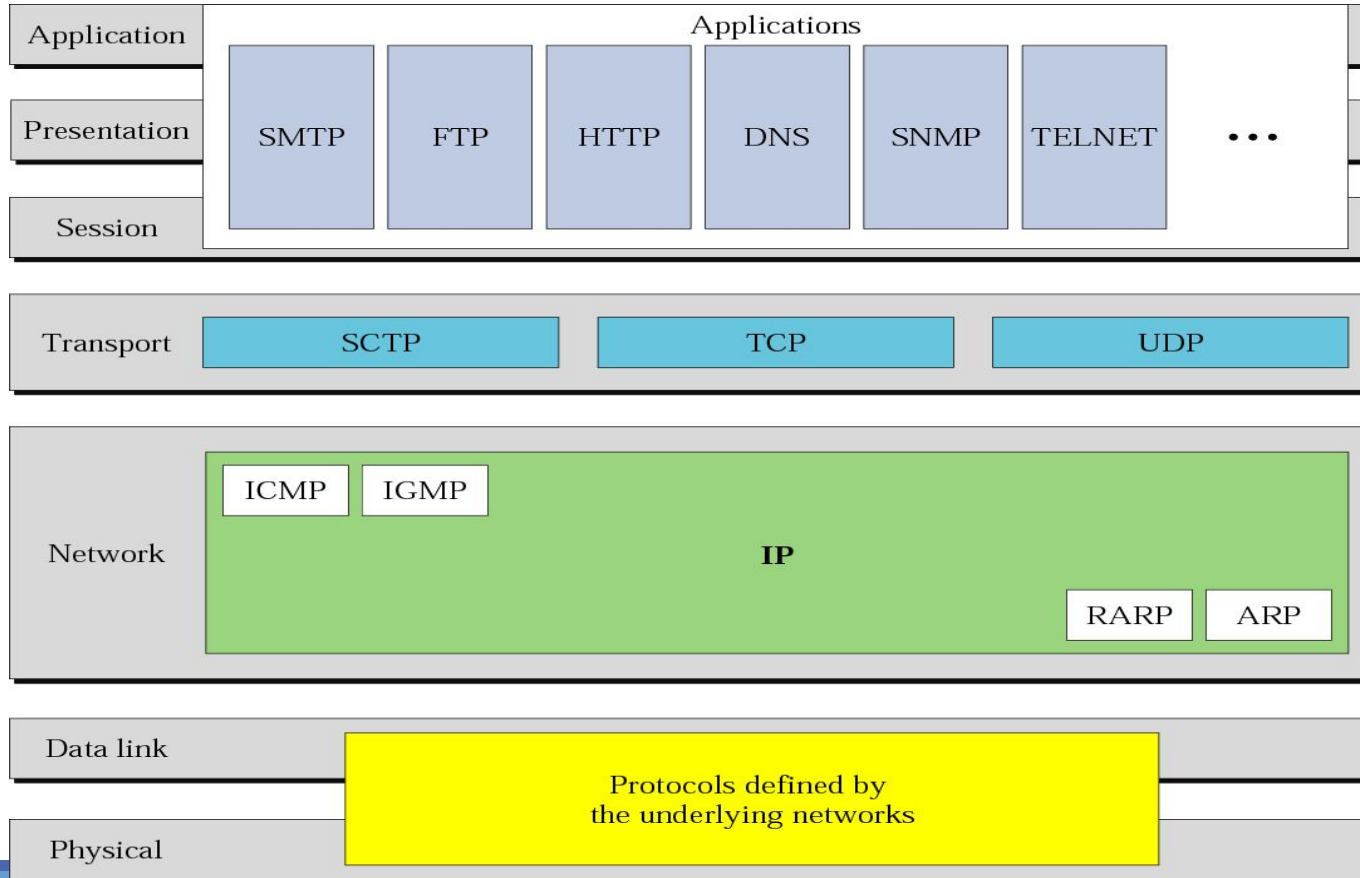
Protocols

- Protocols operate in **layers**.
- A network protocol is essentially a **common language for computers**, enabling devices with different hardware and software to communicate effectively.
- It defines **how data is formatted, transmitted, and processed**, ensuring reliable and standardized communication across networks.
- Protocols prevent data loss, duplication, and misinterpretation, and they also manage error detection, flow control, and session initiation and termination .
- They also provide **security, efficiency, and standardization**, which are essential for modern networks, including the Internet, LANs, and enterprise systems

Key Functions

- **Message Format:** Structure and order of bits in a message
- **Addressing:** How devices identify themselves and the intended recipient
- **Timing:** Sequencing and speed of message exchange
- **Error Detection and Correction:** Identifying and handling corrupted data
- **Session Management:** Initiating, maintaining, and terminating communication sessions

OSI model protocols

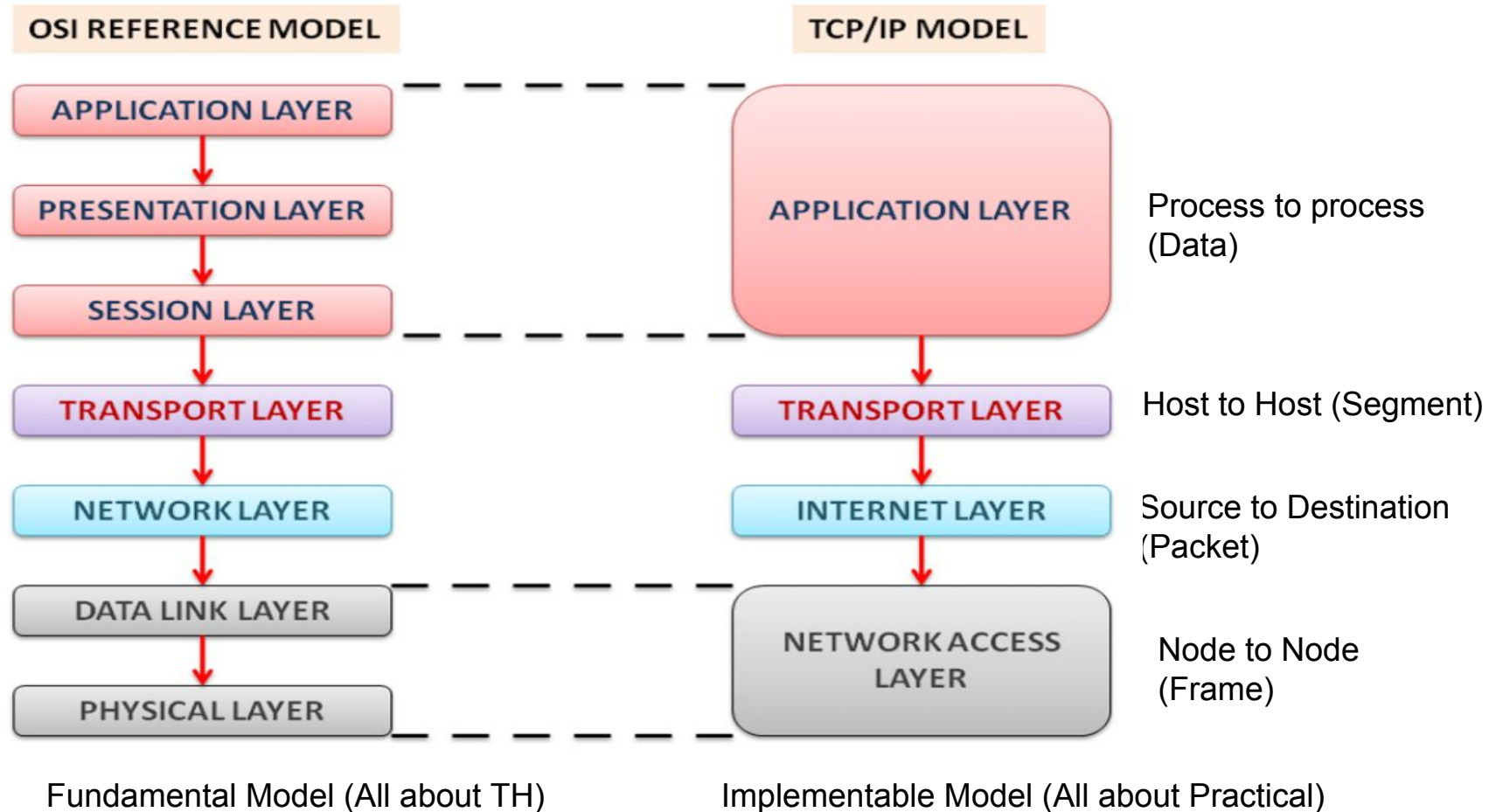


TCP/IP Model

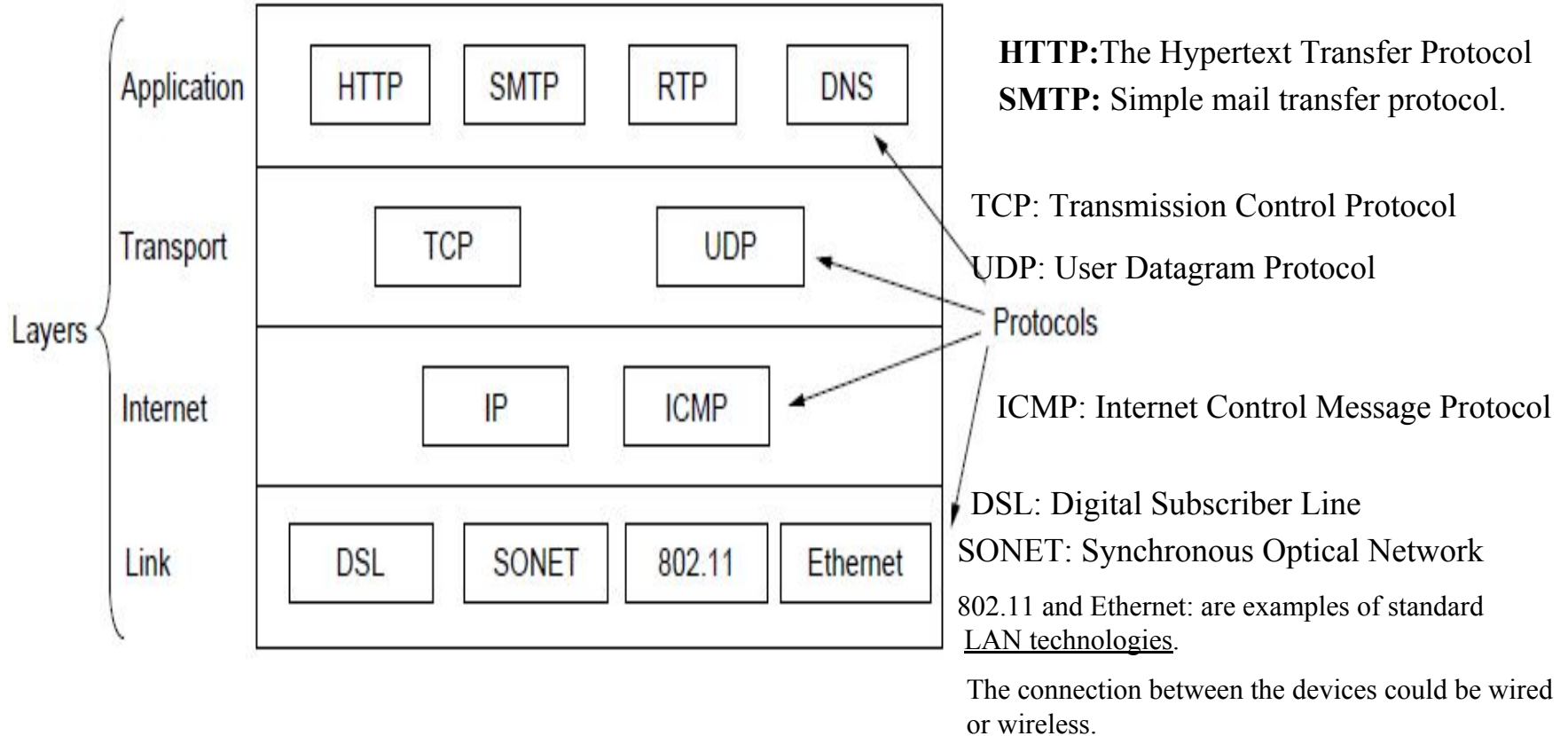
TCP :Transmission Control Protocol, IP : Internet Protocol

- The main work of TCP/IP is to transfer the data of a computer from one device to another.
- TCP/IP model divides its data into packets and combines them at the other end, which helps in maintaining the accuracy of the data while transferring from one end to another end.
- **IP** finds the destination of the mail and TCP has the work to send and receive the mail.

The **Advanced Research Projects Agency Network (ARPANET)** was the first wide-area [packet-switched network](#) with distributed control and one of the first computer networks to implement the [TCP/IP](#) protocol suite.



TCP/IP Model: Protocols



Comparison of OSI & TCP/IP Reference Models

OSI (Open System Interconnection)

developed by ISO (International Standard Organization)

1. OSI is a generic, protocol independent standard, acting as a communication gateway between the network and end user.
2. In OSI model the transport layer **guarantees** the delivery of packets.
3. Follows **vertical approach**.
4. OSI model has a **separate** Presentation layer and Session layer.
5. Transport Layer is **Connection Oriented**.

TCP/IP(Transmission Control Protocol / Internet Protocol) developed by ARPANET (Advanced Research Project Agency Network).

1. TCP/IP model is based on **standard protocols** around which the Internet has developed. It is a communication protocol, which allows connection of hosts over a network.
2. In TCP/IP model the transport layer **does not guarantees** delivery of packets. Still the TCP/IP model is more reliable.
3. Follows **horizontal approach**.
4. TCP/IP **does not have** a separate Presentation layer or Session layer.
5. Transport Layer is **both Connection Oriented and Connection less**.



Comparison of OSI & TCP/IP Reference Models

OSI (Open System Interconnection)

6. **Network Layer** is both Connection Oriented and Connection less.

7. OSI is a reference model around which the networks are built. Generally it is used as a **guidance tool**.

8. Protocols are hidden in OSI model and are easily replaced as the technology changes.

9. OSI model has a problem of fitting the protocols into the model.

10. OSI model defines services, interfaces and protocols very clearly and makes clear distinction between them. It is **protocol independent**.

11. It has **7 layers**

TCP/IP (Transmission Control Protocol / Internet Protocol)

6. Network Layer is **Connection less**.

7. TCP/IP model is, in a way implementation of the OSI model.

8. In TCP/IP **replacing protocol is not easy**.

9. TCP/IP model does not fit any protocol

10. In TCP/IP, services, interfaces and protocols are not clearly separated. It is also **protocol dependent**.

11. It has **4 layers**

Ethernet Standards

Comparison of Baseband and Broadband transmission (only for information)

Features	Baseband Transmission	Broadband Transmission
Definition	It is a data transmission technique in which one signal needs the whole bandwidth of the channel to transfer the data.	It is a transmission technology in which many signals with different frequencies send data across a single channel at the same time.
Signal Type	It utilizes digital signals.	It utilizes analog signals.
Topology	It operates with bus topology.	It operates with both bus and tree topology.
Number of Channels	It utilizes the same channel for sending and receiving data.	It utilizes two channels, one for transmission and the second for data reception.
Distance Covered	Signals are only capable of travelling limited distances.	Signals may be transmitted across long distances without attenuation.
Installation	It is simple and easy to install and maintain.	It is complex to install and maintain.
Cost	It is less expensive to design.	It is costly to design.
Encoding Technique	Manchester and differential Manchester encoding are used in baseband.	It utilizes the PSK (Phase shift keying) encoding.
Transfer medium	It utilizes coaxial cables, wires, and twisted-pair cables as the transfer medium for digital signals.	It sends digital signals via coaxial cable, optical fibre cables, and radio waves.
Application	It is usually found in Ethernet.	It is usually found in telephone networks and cables.

Ethernet Standards

- Ethernet: It is a LAN protocol that is used in Bus and Star topologies and implements CSMA/CD as the medium access method
- Original (traditional) Ethernet developed in 1980 by three companies: Digital, Intel, Xerox (DIX).
- In 1985, the Computer Society of the IEEE started a project, called Project 802, to set standards to enable intercommunication among equipment from a variety of manufacturers. Current version is called IEEE Ethernet
- Basic frame format which is required for all MAC implementation is defined in IEEE

MAC or Ethernet address

- A MAC address (media access control address) is a 12-digit hexadecimal number assigned to each device connected to the network.
- Primarily specified as a unique identifier during device manufacturing, the MAC address is often found on a device's network interface card (NIC).
- MAC addresses are also referred to as Ethernet addresses but multiple types of networks use MAC addressing, including Ethernet, Wi-Fi, and Bluetooth
- MAC addresses are used to identify a device on the network.
- A MAC address is responsible for local identification and an IP address for global identification.

MAC addresses or Ethernet address

- Traditional MAC addresses are 12-digit (6 bytes or 48 bits) hexadecimal numbers
- The leftmost six digits (24 bits), called a prefix, are associated with the adapter manufacturer (M)
- Each vendor registers and obtains MAC prefixes as assigned by the IEEE
- For example, the prefixes 00:13:10, 00:25:9C, and 68:7F:74 (plus others) belong to Linksys (Cisco Systems).
- The rightmost digits of a MAC address represent an identification number for the specific device (S).
- Among all devices manufactured with the same vendor prefix, each is given a unique 24-bit number

MAC addresses or Ethernet address

- Six bytes = 48 bits **06-01-02-01-2C-4B**
- Flat address not hierarchical
- Burned into the NIC ROM

Examples of a unicast and a multicast MAC address

MAC address: 4A:30:10:19:10:1A

First Byte = 4A = 0100 1010

Last bit = 0

Its unicast address

MAC address: 01:00:5E:00:01:01

First Byte = 01 = 0000 0001

Last bit = 1

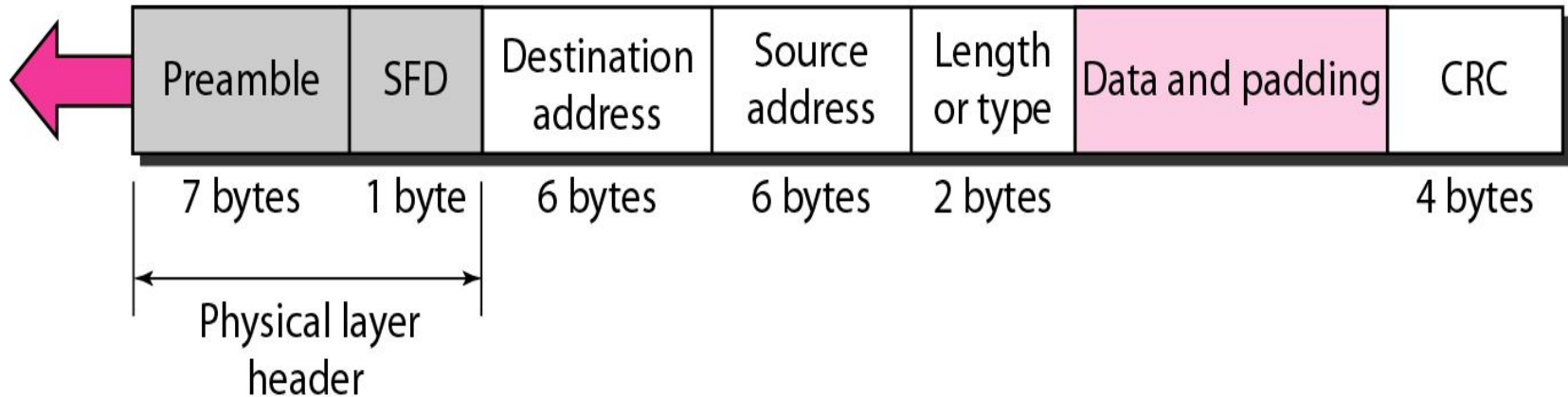
Its multicast address

802.3 MAC frame

Basic frame format which is required for all MAC implementation is defined in **IEEE 802.3 standard**.

Preamble: 56 bits of alternating 1s and 0s.

SFD: Start frame delimiter, flag (10101011)



802.3 MAC frame

PREAMBLE – Ethernet frame starts with a 7-Bytes Preamble. This is a pattern of alternative 0's and 1's which indicates starting of the frame and allow sender and receiver to establish bit synchronization. PRE (Preamble) indicates the receiver that frame is coming

Start of frame delimiter (SFD) – This is a 1-Byte field that is always set to 10101011. SFD indicates that upcoming bits are starting the frame, which is the destination address. The SFD warns station or stations that this is the last chance for synchronization.

Destination Address – This is a 6-Byte field that contains the MAC address of the machine for which data is destined.

Source Address – This is a 6-Byte field that contains the MAC address of the source machine. As Source Address is always an individual address (Unicast), the least significant bit of the first byte is always 0.

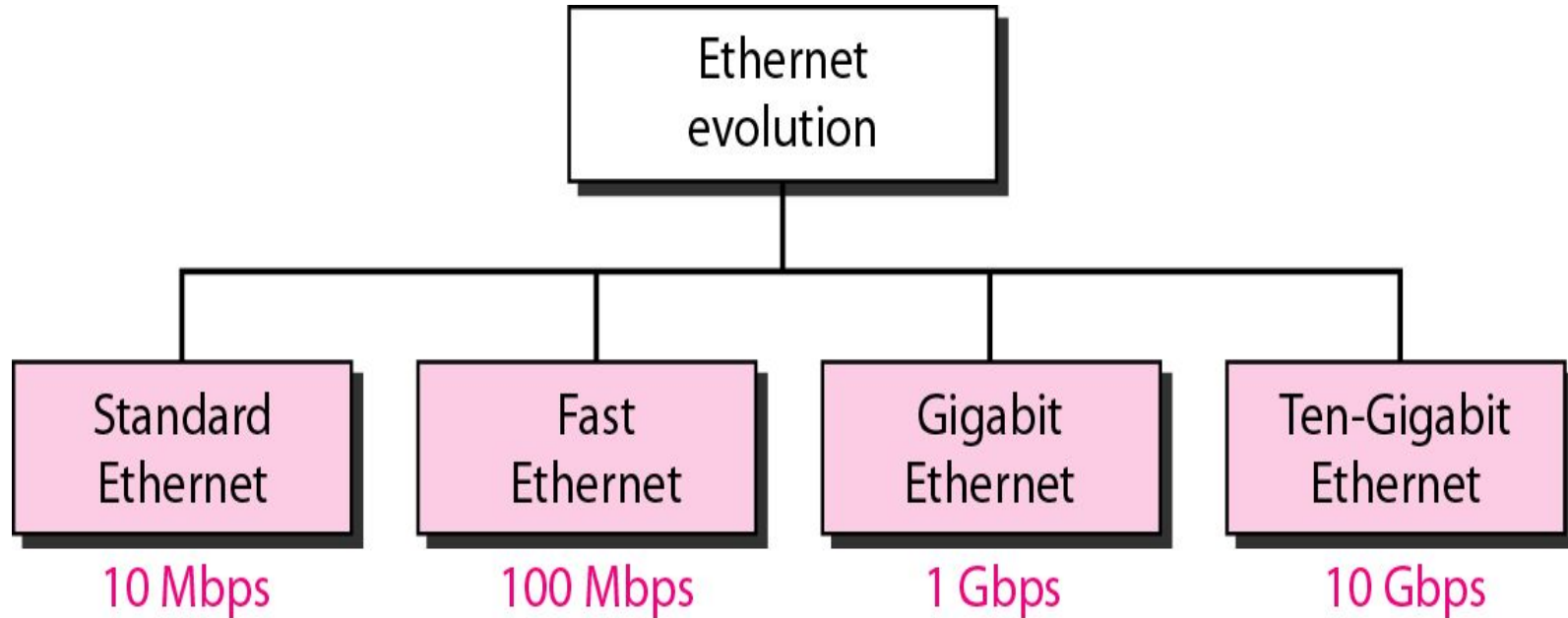
802.3 MAC frame

Length – Length is a 2-Byte field, which indicates the length of the entire Ethernet frame. This 16-bit field can hold a length value between 0 to 65534, but length cannot be larger than 1500 Bytes because of some own limitations of Ethernet.

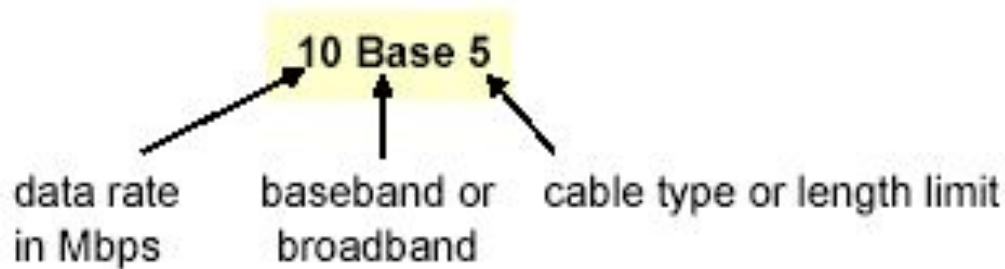
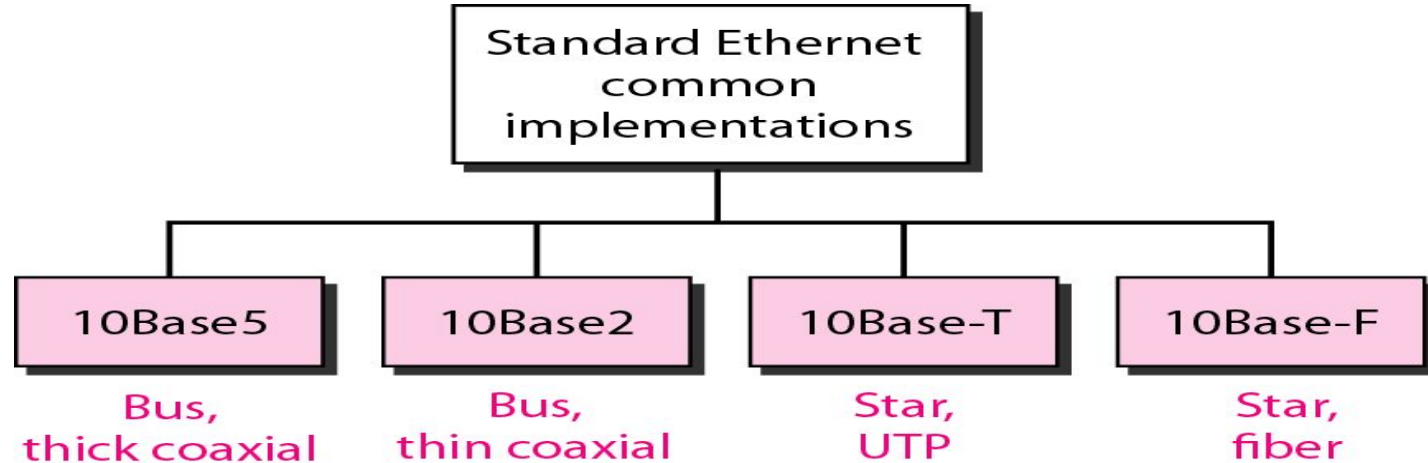
Data – This is the place where actual data is inserted, also known as **Payload**. Both IP header and data will be inserted here if Internet Protocol is used over Ethernet. The maximum data present may be as long as 1500 Bytes.

Cyclic Redundancy Check (CRC) – CRC is 4 Byte field. This field contains a 32-bits hash code of data, which is generated over the Destination Address, Source Address, Length, and Data field. If the checksum computed by destination is not the same as sent checksum value, data received is corrupted.

Ethernet evolution through four generations



Categories of traditional Ethernet



Standard Ethernet topology

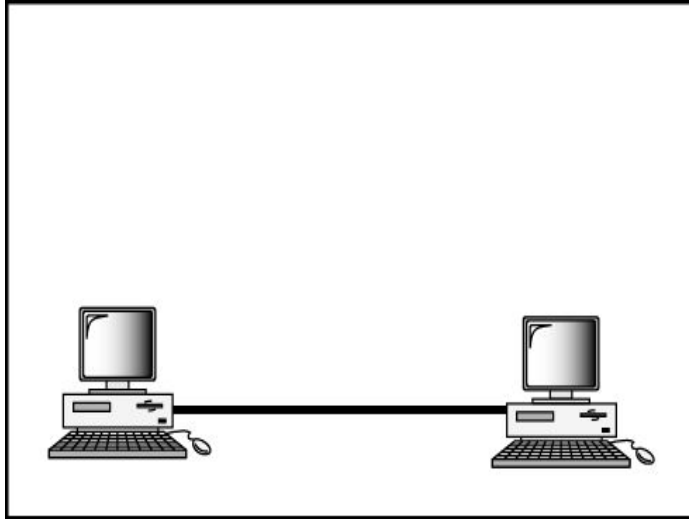
Name	Cable Max.	Max Cable Segment Length	Nodes /segment	Topology
10Base5	thick coax	500 meters	100	Bus
10Base2	thin coax	185 meters	30	Bus
10BaseT	twisted pair	100 meters	1	Star
10BaseF	Fiber Optic	2Km	1	Star

Fast Ethernet

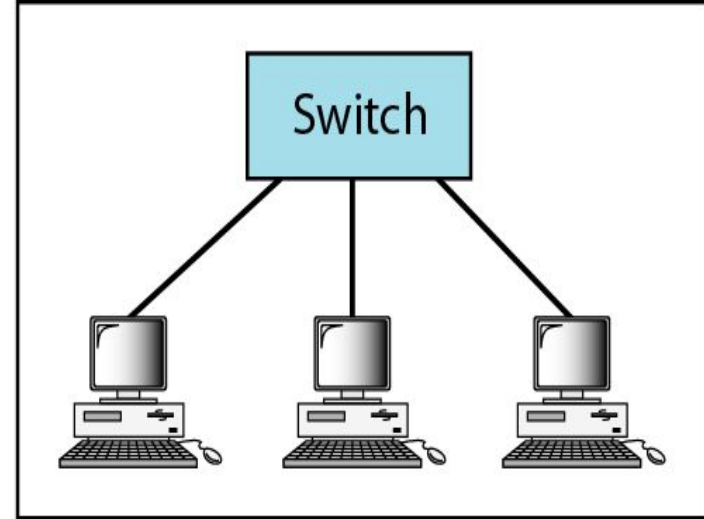
1. Upgrade the data rate to 100 Mbps.
2. Make it compatible with Standard Ethernet.
3. Keep the same 48-bit address.
4. Keep the same frame format
 - media: twisted pair (CAT 5) or fiber optic cable
 - Star-wire topology- Similar to 10BASE-T

Name	Cable	Max. segment	
100Base-T4	Twisted pair	100 m	CAT 3
100Base-TX	Twisted pair	100 m	CAT 5
100Base-FX	Fiber optics	2000 m	

Fast Ethernet Topology



a. Point-to-point



b. Star

Gigabit Ethernet

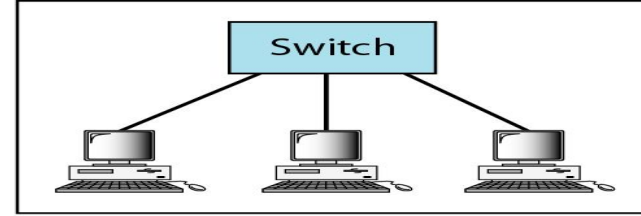
1. Upgrade the data rate to 1 Gbps.
2. Make it compatible with Standard or Fast Ethernet.
3. Use the same 48-bit address.
4. Use the same frame format.
5. Keep the same minimum and maximum frame lengths.
6. Support auto-negotiation as defined in Fast
7. Ethernet Operates in full/half duplex modes mostly full duplex

Name	Cable	Max. segment	Advantages
1000Base-SX	Fiber optics	550 m	Multimode fiber (50, 62.5 microns)
1000Base-LX	Fiber optics	5000 m	Single (10 μ) or multimode (50, 62.5 μ)
1000Base-CX	2 Pairs of STP	25 m	Shielded twisted pair
1000Base-T	4 Pairs of UTP	100 m	Standard category 5 UTP

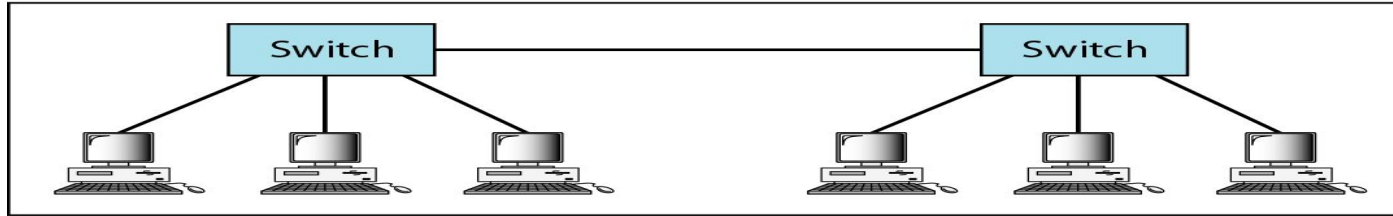
Gigabit Ethernet Topologies



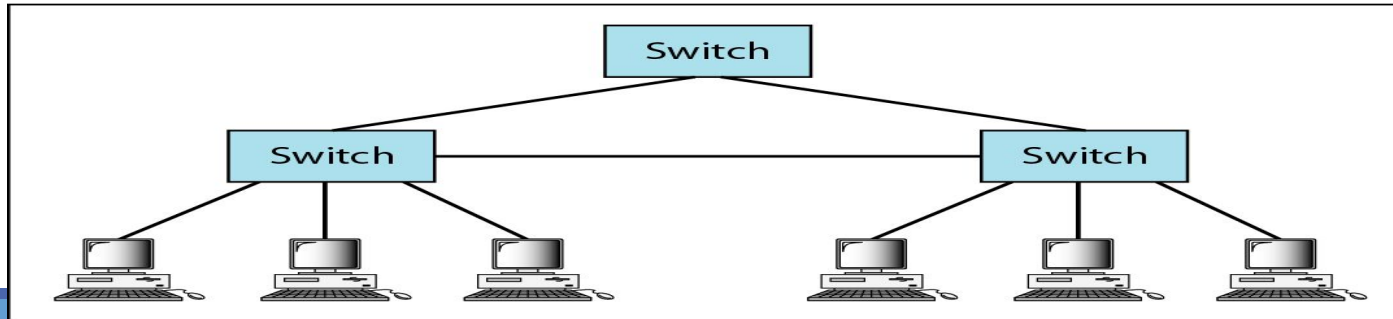
a. Point-to-point



b. Star



c. Two stars



d. Hierarchy of stars

Ethernet Type	Bandwidth	Cable Type	Maximum Distance
10Base-T	10Mbps	Cat 3/Cat 5 UTP	100m
100Base-TX	100Mbps	Cat 5 UTP	100m
100Base-TX	200Mbps	Cat 5 UTP	100m
100Base-FX	100Mbps	Multi-mode Fiber	400m
100Base- FX	200Mbps	Multi-mode Fiber	2Km
1000Base-T	1Gbps	Cat 5e UTP	100m
1000Base-TX	1Gbps	Cat 6 UTP	100m
1000Base-SX	1Gbps	Multi-mode Fiber	550m
1000Base-LX	1Gbps	Single-mode Fiber	2Km
10GBase-T	10Gbps	Cat 6a/Cat 7 UTP	100m
10GBase-LX	10Gbps	Multi-mode Fiber	100m
10GBase-LX	10Gbps	Single-mode Fiber	10Km

Thank You

Any Questions